The Impact of the Distance Between Cycles on Elementary Trapping Sets

Haoran Xiong, *Member, IEEE*, Guanghui Wang, Zhiming Ma, and Guiying Yan

***Abstract*—Elementary trapping sets (ETSs) are the main culprits of the performance of low-density parity-check (LDPC) codes in the error floor region. Due to their large quantities and complex structures, ETSs are difficult to analyze. This paper studies the impact of the distance between cycles on ETSs, focusing on two special graph classes: theta graphs and dumbbell graphs, which correspond to cycles with negative and non-negative distances, respectively. We determine the Turán numbers of these graphs and prove that increasing the distance between cycles can eliminate more ETSs. Additionally, using the linear state-space model and spectral theory, we prove that increasing the length of cycles or distance between cycles decreases the spectral radius of the system matrix, thereby reducing the harmfulness of ETSs. This is consistent with the conclusion obtained using Turán numbers. For specific cases when removing two 6-cycles with distance of -1, 0 and 1, respectively, we calculate the sizes, spectral radii, and error probabilities of ETSs. These results confirm that the performance of LDPC codes improves as the distance between cycles increases. Furthermore, we design the PEG-CYCLE algorithm, which greedily maximizes the distance between cycles in the Tanner graph. Numerical results show that the QC-LDPC codes constructed by our method achieve performance comparable to or even superior to state-of-the-art construction methods.**

***Index Terms*—Low-density parity-check (LDPC) code, elementary trapping set (ETS), Turán number, spectral radius.**

I. Introduction

As an important class of modern coding theory, low-density parity-check (LDPC) codes have received widespread attention for their excellent error-correction capabilities and efficient parallel iterative decoding algorithm [1], [2], [3]. As capacity-approaching codes, LDPC codes are extensively used in various systems, such as Wi-Fi, optical communication, microwave systems, and data storage [4]. Among them, quasi-cyclic LDPC (QC-LDPC) codes are particularly important due to their efficient hardware implementation, making them a popular choice in standards such as 5G NR [5], [6], [7].

The error rate curve of LDPC codes under iterative decoding is typically divided into two regions: the waterfall region and the error floor region. The error floor region occurs at high signal-to-noise ratio (SNR) and is characterized by a slower error rate reduction.

A Tanner graph, which is a bipartite graph composed of variable nodes and check nodes, corresponds one-to-one to an LDPC code and plays an important role in the decoding process [2], [8]. A Tanner graph is called variable-regular if all its variable nodes have the same degree, denoted by γ.

The error floor behavior of an LDPC code is primarily caused by specific graph structures in the Tanner graph, known as trapping sets [9]. An (a, b) trapping set (TS) is a subgraph of the Tanner graph which is induced by a variable nodes in the set and their check node neighbors, with b check nodes of odd degrees and an arbitrary number of even degree check nodes. Here, a is called the size of the TS. An elementary trapping set (ETS) is a specific type of TS where all check nodes have degrees of either 1 or 2 and is considered the most harmful among TSs [10], [11].

Removing certain ETSs in the Tanner graph is important for improving the performance of LDPC codes. However, identifying the non-isomorphic structures of ETSs with varying values of a and b is challenging, and determining the minimum size of ETSs is NP-hard [12]. Thus, researchers focused on structural properties that directly influence ETSs. The most widely studied approaches include optimizing the minimum distance d_{min} [13], [14], [15], [16] and increasing the girth g of the Tanner graph [17], [18], [19], [20], [21], [22], [23]. Additionally, other methods eliminated ETSs by removing specific graph structures [24], [25], [26], [27], [28], [29], [30], [31].

While increasing girth helps eliminate trapping sets and improves performance, it requires longer code lengths for fixed variable node degrees. Furthermore, the girth of QC-LDPC codes is bounded above by 12 [32]. These constraints limit the effectiveness of girth optimization for performance enhancement. While the size of trapping sets can effectively characterize the harmfulness of substructures in LDPC codes, its practical implementation remains challenging due to their large quantity, intricate structures, and high computational complexity. To address these limitations, in this paper, we introduce a new structural property: the distance between

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cycles, for analyzing and eliminating ETSs. This idea not only complements existing techniques but also offers a more effective way to improve the performance of LDPC codes in the error floor region.

A. Previous Work

Hashemi and Banihashemi substantiated that small ETSs can be generated by short cycles or non-cycle graphs, which characterize the basic structures of ETSs [28], [29]. Amirzade et al. [30] constructed girth-6 QC-LDPC codes by avoiding 8-cycles with a chord, eliminating all (a, b)-ETSs with $a \leq 5$, $b \leq 3$ for $\gamma = 3$, and $a \leq 7$, $b \leq 4$ for $\gamma = 4$. Furthermore, they derived an inequality $b \geq a\gamma - \frac{2a^3}{4a-3}$ for (a, b)-ETSs in the Tanner graphs with a variable-regular degree γ. In [31], the authors improved this bound and considered the case of girth 8. By limiting common edges between 8-cycles, they eliminated (a, b)-ETSs with $a \leq 7$, $b < 3$ when $\gamma = 3$ and $a \leq 9$, $b < 6$ when $\gamma = 4$. We observe that removing 8-cycles with a chord [30] and restricting common edges between 8-cycles [31] imply the elimination of cycles with small distances.

B. Our Contributions

In this paper, we focus on ETSs in variable-regular Tanner graphs, using the variable node (VN) graphs [33], which bijectively represent ETSs. We analyze two special classes of graphs—theta graphs and dumbbell graphs—which form the fundamental structures of VN graphs. We establish the relationship between the removal of specific substructures in the VN graph and constraints on ETSs in the Tanner graph through Turán numbers. As a cornerstone of graph theory, Turán numbers provide profound insights into how local constraints influence global graph structures. From the perspective of Turán numbers, we prove that removing structures with smaller Turán numbers in the VN graph can eliminate more ETSs.

To further analyze the behavior of these structures in ETSs, we use the linear state-space model, which is a tool for analyzing the message-passing process within ETSs [34]. By spectral theory (eigenvalue analysis of matrices), we prove that increasing cycle lengths or cycle distances reduces the spectral radius of the corresponding system matrix. This conclusion is supported by numerical examples. Combined with the spectral radius analysis in [34], we conclude that increasing cycle lengths or distances improves the performance of LDPC codes, which is consistent with the conclusion obtained from Turán numbers. Furthermore, we utilize the linear state-space model to simulate the rate of error probability reduction of ETSs, showing that structures with larger cycle distances exhibit a faster reduction rate.

Based on these results, we propose the PEG-CYCLE algorithm, which greedily maximizes cycle distances during the construction of the Tanner graph. This algorithm is applicable to both LDPC and QC-LDPC codes, including regular, irregular, fully connected, and non-fully connected cases. Simulations show that the codes constructed using our algorithm achieve performance comparable to or even better than those generated by state-of-the-art construction methods.

Although the methods to eliminate ETSs by removing special substructures, such as [25], [26], [30], [31], and [41], have implicitly suggested that cycles in Tanner graphs should not be too dense, these papers focus on the impact of removing specific substructures. They neither explicitly proposed the parameter of cycle distance nor provided theoretical analysis on increasing the distance between cycles. Our work introduces a novel approach for removing small ETSs in the Tanner graph by increasing the distance between cycles, thereby improving the performance of LDPC codes in the error floor region.

C. Paper Outline

The structure of this paper is as follows: Section II introduces essential definitions and notations. Section III presents theoretical results from two perspectives: Turán numbers of specific graphs and the spectral radius of the system matrix. In Subsection III-A, we determine the Turán numbers of specific dumbbell graphs and extend the results to general cases. In Subsection III-B, we use spectral theory to analyze the variation of spectral radius as cycle lengths and distances change. Section IV examines the case of two 6-cycles with distances −1, 0, and 1, and calculates the sizes, spectral radii and the rate of error probability reduction of corresponding ETSs. Based on these findings, Section V designs the PEG-CYCLE algorithm and presents simulation results. Section VI offers conclusion of this paper.

II. Definitions and Preliminaries

In this paper, matrices are represented by bold capital letters (e.g., $\mathbf{A}$), and vectors by bold lowercase letters (e.g., $\boldsymbol{x}$). Their entries are denoted as $\mathbf{A}(i, j)$ for matrices and $\boldsymbol{x}(i)$ for vectors. Scalars or variables are represented by lowercase letters (e.g., k).

A. Graph Theory

An undirected graph $G = (V(G), E(G))$ consists of a non-empty vertex set $V(G)$ and an edge set $E(G)$, where edges are unordered pairs of vertices. Two vertices u and v in $V(G)$ are adjacent, or neighbors, if they are connected by an edge, denoted as $uv \in E(G)$. For an edge uv, we denote by G_{uv} the graph obtained by subdividing the edge uv, which means introducing a new vertex on the edge uv.

The neighborhood of a vertex v is $N_G(v) = \{u \in V(G) \mid uv \in E(G)\}$. The degree of a vertex $v \in G$ is $d_G(v) = |N_G(v)|$. A graph G is regular if all its vertices have the same degree. For brevity, we use V, E, $N(v)$, and $d(v)$ when no ambiguity arises. Let $\delta(G)$ denote the minimum degree of G. A graph without loops or multiple edges is a simple graph, which is the focus of this paper.

A simple graph is complete, denoted K_n, if there is an edge connecting every pair of vertices. A simple graph is bipartite if its vertices can be partitioned into two sets V_1, V_2 such that no edge joins two vertices in the same set. Additionally, a bipartite graph is complete, if every vertex in V_1 is connected to every vertex in V_2, and is denoted $K_{m,n}$ with m and n vertices in each part, respectively.

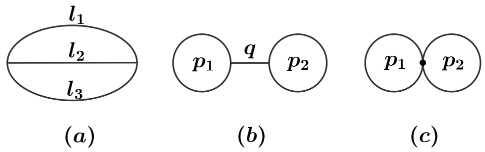

Fig. 1. (a) Theta graph $\theta(l_1, l_2, l_3)$; (b)-(c) Dumbbell graphs $D_b(r_1, r_2; q)$ for $q \geq 1$ and $q = 0$.

Given a subset $S \subseteq V$, the neighborhoods of S are denoted by $N(S)$. The induced subgraph generated by S is defined as $G[S] = (S, E(G[S]))$, where $E(G[S]) = \{uv \in E(G) \mid u, v \in S\}$. We denote by $G-S$ the graph obtained from G by removing all vertices in S along with all edges incident to them. For a subset $E' \subseteq E(G)$, $G - E'$ denotes the graph obtained by removing the edges in E' from G, while keeping $V(G)$ unchanged.

A path of length $k-1$, denoted by P_k, is a sequence of distinct vertices $v_1 v_2 \ldots v_k$, where $v_i v_{i+1} \in E$ for all $1 \leq i \leq k-1$. Two or more paths are internally disjoint if they share only endpoints. A graph G is connected if there exists a path between any two distinct vertices. A cycle of length k, denoted by C_k, is a path P_k with an additional edge connecting the first and last vertices.

In this paper, we focus on the impact of distance between cycles on the performance of LDPC codes. To quantify the distance between two cycles, we introduce the following special graphs in graph theory, as shown in Fig. 1.

Definition 1: A theta graph $\theta(l_1, l_2, l_3)$ is the graph formed by three internally disjoint paths with common endpoints, where the three paths have lengths l_1, l_2, and l_3, respectively.

Definition 2: A dumbbell graph $D_b(r_1, r_2; q)$ consists of two vertex-disjoint cycles, C_{r_1} and C_{r_2},

and a path P_{q+1} of length q ($q \geq 1$) with sharing only its endpoints with the two cycles. $D_b(r_1, r_2; 0)$ represents that C_{r_1} intersects C_{r_2} at exactly one common vertex.

Remark 1: Assume $l_1 \leq l_2 \leq l_3$ for $\theta(l_1, l_2, l_3)$ and $r_1 \leq r_2$ for $D_b(r_1, r_2; q)$ to avoid repetitive discussions about isomorphic graphs. As there are no multiple edges, then $l_2 \geq 2$ and r_1, $r_2 \geq 3$.

For the dumbbell graph $D_b(r_1, r_2; q)$, the distance between C_{r_1} and C_{r_2} can be naturally defined as q, the length of the path connecting them. In contrast, theta graphs are used to describe cases where two cycles share common edges, and their distance is defined as a negative number. Note that a theta graph $\theta(l_1, l_2, l_3)$ can describe the following three cases:

- $C_{l_1+l_2}$ and $C_{l_1+l_3}$ with a distance of $-l_1$;
- $C_{l_1+l_2}$ and $C_{l_2+l_3}$ with a distance of $-l_2$;
- $C_{l_1+l_3}$ and $C_{l_2+l_3}$ with a distance of $-l_3$.

To avoid unnecessary discussion of isomorphic graphs, we measure the distance using the shortest path in a theta graph. Therefore, $\theta(l_1, l_2, l_3)$ represents $C_{l_1+l_2}$ and $C_{l_1+l_3}$ with a distance of $-l_1$.

Based on the above discussion, we define the distance between the cycles represented by the two graphs as follows:

Definition 3: The dumbbell graph $D_b(r_1, r_2; q)$ represents a cycle of length r_1 and a cycle of length r_2 with distance q. The theta graph $\theta(l_1, l_2, l_3)$ represents a cycle of length $l_1 + l_2$ and a cycle of length $l_1 + l_3$ with distance $-l_1$.

Since $l_1 \leq l_2 \leq l_3$, then $l_1 \leq \frac{l_1+l_2}{2}$ and $l_1 \leq \frac{l_1+l_3}{2}$. Therefore, for two cycles of lengths of c_1 and c_2, the distance d satisfies $d \geq \max\left\{-\left\lfloor \frac{c_1}{2} \right\rfloor, -\left\lfloor \frac{c_2}{2} \right\rfloor\right\}$. Then we can map different combinations of cycle lengths and distances to theta graphs or dumbbell graphs. To show the impact of cycle distances on ETSs, we introduce the following definitions from extremal graph theory.

Definition 4: Let H be a fixed graph. The Turán number $ex(n, H)$ is the maximum number of edges in any graph of n vertices that contains no subgraph isomorphic to H. A graph G on n vertices with $ex(n, H)$ edges and without a copy of H is called an extremal graph for H.

The earliest result about Turán numbers can be traced back to Mantel's theorem from 1907, which is stated as follows:

Theorem 1 (Mantel's theorem, [35]): For all $n \geq 3$, $ex(n, C_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$.

B. Spectra of Graphs

A directed graph (or digraph) $D = (V, A)$ consists of a set of vertices V and a set of arcs (directed edges) A. An arc (u, v) is directed from u (the head) to v (the tail). A digraph is simple if it contains no loops or parallel arcs.

The adjacency matrix $\mathbf{A}(D)$ of a digraph D is a $|V| \times |V|$ matrix, where the (i, j) entry represents the number of arcs from v_i to v_j.

Let $\mathbf{M}$ be a real $n \times n$ matrix with nonnegative entries. $\mathbf{M}$ is called irreducible if for all i, j, there exists a positive integer k such that $(\mathbf{M}^k)(i, j) > 0$. For a matrix $\mathbf{A}$, we write $\mathbf{A} > 0$ (or $\mathbf{A} \geq 0$) to indicate that all entries of $\mathbf{A}$ are positive (nonnegative). Similarly, for two matrices $\mathbf{A}_1$ and $\mathbf{A}_2$, we use $\mathbf{A}_1 \geq \mathbf{A}_2$ to mean that $\mathbf{A}_1 - \mathbf{A}_2 \geq 0$. The same notation applies to vectors. The spectral radius $\rho(\mathbf{M})$ of $\mathbf{M}$ is defined as the maximum modulus of its eigenvalues:

$$\rho(\mathbf{M}) = \max\{|\lambda| : \lambda \text{is an eigenvalue of } \mathbf{M}\}.$$

We introduce the following theorem about the eigenvalues of irreducible matrices, which is one of the most important results in spectral theory.

Theorem 2 (Perron-Frobenius Theorem [36]): Let $\mathbf{M} \geq 0$ be an $n \times n$ irreducible matrix. Then the spectral radius of $\mathbf{M}$ ρ satisfies the following properties:

(i) There exists a real vector $\boldsymbol{x_0} > 0$ such that $\mathbf{M}\boldsymbol{x_0} = \rho \boldsymbol{x_0}$.

(ii) If $\boldsymbol{x} \geq 0$, $\boldsymbol{x} \neq 0$ and $\mathbf{M}\boldsymbol{x} \leq \rho' \boldsymbol{x}$, then $\boldsymbol{x} > 0$ and $\rho \leq \rho'$. Moreover, $\rho = \rho'$ if and only if $\mathbf{M}\boldsymbol{x} = \rho' \boldsymbol{x}$.

(iii) The spectral radius ρ satisfies the inequality:

$$\min_i \sum_{j=1}^{n} \mathbf{M}(i, j) \leq \rho \leq \max_i \sum_{j=1}^{n} \mathbf{M}(i, j).$$

One of the equalities holds if and only if $\min\limits_i \sum_{j=1}^{n} \mathbf{M}(i, j) = \max\limits_i \sum_{j=1}^{n} \mathbf{M}(i, j)$.

C. Coding Theory

A Tanner graph $G = (L \cup R, E)$ is a bipartite graph corresponding to the parity-check matrix $\mathbf{H}$ of an LDPC code, where L represents variable nodes, corresponding to columns of $\mathbf{H}$, and R represents check nodes, corresponding to rows. An edge connects the i-th check node to the j-th variable node if the entry $\mathbf{H}(i, j) = 1$. Specifically, the graph is variable-regular if every variable node has the same degree γ.

For a lifting degree p, the parity-check matrix $\mathbf{H}$ of a QC-LDPC code can be represented by $p \times p$ circulant permutation matrices as follows [32]:

$$\mathbf{H} = \begin{bmatrix} \mathbf{I}^{(p_{1,1})} & \mathbf{I}^{(p_{1,2})} & \dots & \mathbf{I}^{(p_{1,\eta})} \\ \mathbf{I}^{(p_{2,1})} & \mathbf{I}^{(p_{2,2})} & \dots & \mathbf{I}^{(p_{2,\eta})} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{I}^{(p_{\gamma,1})} & \mathbf{I}^{(p_{\gamma,2})} & \dots & \mathbf{I}^{(p_{\gamma,\eta})} \end{bmatrix} \tag{1}$$

where $\mathbf{I}^{(p_{i,j})}$ represents a $p \times p$ circulant permutation matrix (CPM) with lifting value $p_{i,j} \in \{0, 1, 2, \dots, p-1, \infty\}$. Specifically, $\mathbf{I}^{(\infty)}$ corresponds to a $p \times p$ zero matrix, while for $0 \le r \le p-1$, $\mathbf{I}^{(p_{i,j})}$ has a '1' at position $(r, (r + p_{i,j}) \mod p)$ and '0's elsewhere.

A QC-LDPC code is fully connected if $\mathbf{H}$ contains no $\mathbf{I}^{(\infty)}$ entry; otherwise, it is non-fully connected. Two auxiliary matrices are defined: the exponent matrix $\mathbf{E} = (p_{i,j})$ and the base matrix $\mathbf{B} = (b_{i,j})$, where $b_{i,j} = 1$ if $p_{i,j} \neq \infty$ and 0 otherwise.

For a subset $S \subseteq L$, an (a, b) trapping set (TS) is an induced subgraph $G[S \cup N(S)]$ with $|S| = a$, and b is the number of odd-degree vertices in $N(S)$. An elementary trapping set (ETS) is a special type of TS in which all the check nodes have degrees of either 1 or 2.

The variable node (VN) graph $G_{VN} = (V_{VN}, E_{VN})$ of an ETS has vertices of the variable nodes, while E_{VN} corresponds to the pairs of variable nodes connected through degree-2 check nodes [33]. In a variable-regular Tanner graph with degree γ, there exists a one-to-one correspondence between an (a, b)-ETS and its corresponding VN graph. Therefore, in this paper, we equate the VN graph with its corresponding ETS.

For a VN graph $G_{VN} = (V_{VN}, E_{VN})$ of an (a, b)-ETS in a variable-regular Tanner graph with degree γ, we have $|V_{VN}| = a$ and $|E_{VN}| = \frac{1}{2}(a\gamma - b)$. Since $|E_{VN}|$ is an integer, this implies:

- when γ is odd, a and b have the same parity;
- when γ is even, b must be even.

Moreover, if the girth of the Tanner graph exceeds 4, the VN graph of its ETS is simple.

We introduce the linear state-space model in [34], which characterizes the decoding behavior of an ETS. This model is expressed by the following equations:

$$\boldsymbol{x}^{(0)} = \mathbf{B}\boldsymbol{\lambda}; \tag{2a}$$

$$\boldsymbol{x}^{(l)} = \mathbf{A}_{sys}\boldsymbol{x}^{(l-1)} + \mathbf{B}\boldsymbol{\lambda} + \mathbf{B}_{\mathbf{ex}}\boldsymbol{\lambda_{ex}}^{(l)} \quad \text{for } l \ge 1; \tag{2b}$$

$$\tilde{\boldsymbol{\lambda}}^{(l)} = \mathbf{C}\boldsymbol{x}^{(l-1)} + \boldsymbol{\lambda} + \mathbf{D}_{\mathbf{ex}}\boldsymbol{\lambda_{ex}}^{(l)} \quad \text{for } l \ge 1, \tag{2c}$$

where:

- $\boldsymbol{x}^{(l)}$ represents the log-likelihood ratio (LLR) messages from variable nodes to degree-2 check nodes at iteration l.
- $\boldsymbol{\lambda}$ is the intrinsic information from the channel.
- $\boldsymbol{\lambda_{ex}}^{(l)}$ is the messages from the degree-1 check nodes at iteration l.
- $\tilde{\boldsymbol{\lambda}}^{(l)}$ represents the soft-output decisions (LLR) at iteration l.

For an (a, b)-ETS with a variable-regular degree γ, there are $m = a\gamma - b$ edges connecting variable nodes and degree-2 check nodes. The vectors $\boldsymbol{\lambda}$ and $\tilde{\boldsymbol{\lambda}}^{(l)}$ have a entries, while $\boldsymbol{\lambda_{ex}}^{(l)}$ has b entries.

The $m \times m$ system matrix $\mathbf{A}_{sys}$ describes message updates between iterations: $\mathbf{A}_{sys}(i, j) = 1$ indicates message on edge j at iteration $l-1$ contributes to edge i at iteration l. Diagonal elements of $\mathbf{A}_{sys}$ are 0, and the i-th row sums equal $d(v_i) - 1$, where v_i is the head of edge i. Additionally, $\mathbf{A}_{sys}{}^T$ represents the adjacency matrix of the directed graph describing edge-message updates.

The matrices $\mathbf{B}$, $\mathbf{B}_{\mathbf{ex}}$, $\mathbf{C}$, and $\mathbf{D}_{\mathbf{ex}}$ describe relationships between channel information, messages from degree-1 check nodes, messages on the edges, and soft-output decisions. Specifically:

- The $m \times a$ matrix $\mathbf{B}$ maps channel information to the messages on the edges.
- The $m \times b$ matrix $\mathbf{B}_{\mathbf{ex}}$ maps messages from degree-1 check nodes to the messages on the edges.
- The $a \times m$ matrix $\mathbf{C}$ maps the messages on the edges to the soft-output decisions.
- The $a \times b$ matrix $\mathbf{D}_{\mathbf{ex}}$ maps the messages from degree-1 check nodes to the soft-output decisions.

Equations (2a)–(2c) describe the message-passing process:

- Equation (2a): In the initial stage, variable nodes send the intrinsic channel information to degree-2 check nodes.
- Equation (2b): In iteration l, the messages transmitted from each variable node to degree-2 check nodes depend on three components: (1) the messages received from other degree-2 check nodes in iteration $l-1$, (2) the intrinsic channel information, and (3) the messages from degree-1 check nodes.
- Equation (2c): In iteration l, the soft-output decision at each variable node is the sum of three components: (1) the messages received from degree-2 check nodes, (2) the intrinsic channel information, and (3) the messages from degree-1 check nodes.

For a more detailed and complete discussion, readers can refer to [34]. We use the following example in [34] to further explain this model.

Example 1: For a $(5, 3)$-ETS with $\gamma = 3$ shown in *Fig.* 2 (a), we provide explicit expressions for each matrix as follows:

$$\mathbf{A}_{\text{sys}} = \begin{bmatrix} 0&0&0&1&0&0&0&0&1&0&0&0 \\ 0&0&0&1&0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&0&0&1&0&0&1 \\ 0&0&0&0&0&1&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1&0&0&0&0 \\ 0&0&0&0&1&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&1&0&0&0&1&0 \\ 0&1&0&0&0&0&1&0&0&0&0&0 \\ 1&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&1&0&0 \end{bmatrix}$$

TABLE I
SUMMARY OF PARAMETERS AND POTATIONS

$G=(V,E)$	an undirected graph G with vertex set V and edge set E
$D=(V,A)$	a directed graph D with vertex set V and arc set A
G_{uv}	the graph obtained from G by subdividing the edge uv
$d(v)$	degree of a vertex v
$\delta(G)$	the minimum degree of G
$G[S]$	the induced subgraph generated by $S \subseteq V$
K_n	a complete graph with n vertices
$K_{m,n}$	a complete bipartite graph with m and n vertices in each part
P_k	a path of length $k-1$
C_k	a cycle of length k
$\theta(l_1,l_2,l_3)$	a theta graph where the three paths are of lengths l_1, l_2, and l_3
$D_b(r_1,r_2;q)$	a dumbbell graph consisting of two vertex-disjoint cycles, C_{r_1} and C_{r_2}, and a path P_{q+1}
$ex(n,H)$	Turán number of H
$\mathbf{A}(D)$	the adjacency matrix of a digraph D
$\rho(\mathbf{M})$	the spectral radius of a matrix $\mathbf{M}$
γ	the degree of variable nodes in a variable-regular Tanner graph
g	the girth of Tanner graph
p	the lifting degree of a QC-LDPC code
$\mathbf{I}^{(p_{i,j})}$	a circulant permutation matrix with lifting value $p_{i,j}$
$\mathbf{A_{sys}}$	the system matrix of an ETS

$$\mathbf{B}=\begin{bmatrix} 1&0&0&0&0\\ 1&0&0&0&0\\ 1&0&0&0&0\\ 0&1&0&0&0\\ 0&1&0&0&0\\ 0&0&1&0&0\\ 0&0&1&0&0\\ 0&0&0&1&0\\ 0&0&0&1&0\\ 0&0&0&1&0\\ 0&0&0&0&1\\ 0&0&0&0&1 \end{bmatrix} \mathbf{B}_{\text{ex}}=\begin{bmatrix} 0&0&0\\ 0&0&0\\ 0&0&0\\ 1&0&0\\ 1&0&0\\ 0&1&0\\ 0&1&0\\ 0&0&0\\ 0&0&0\\ 0&0&0\\ 0&0&1\\ 0&0&1 \end{bmatrix}$$

$$\mathbf{C}=\begin{bmatrix} 0&0&0&1&0&0&0&0&1&0&0&1\\ 0&0&1&0&0&1&0&0&0&0&0&0\\ 0&0&0&0&1&0&0&1&0&0&0&0\\ 0&1&0&0&0&0&1&0&0&0&1&0\\ 1&0&0&0&0&0&0&0&0&1&0&0 \end{bmatrix}$$

$$\mathbf{D}_{\text{ex}}=\begin{bmatrix} 0&0&0\\ 1&0&0\\ 0&1&0\\ 0&0&0\\ 0&0&1 \end{bmatrix}$$

Note that $\mathbf{A}_{sys}^T$ corresponds to the adjacency matrix of *Fig. 2 (b)*, where $\mathbf{A}_{sys}(i,j)=1$ indicates the edge j participates in the update of the edge i (e.g., edges 4 and 9 update edge 1).

In this linear state-space model, the spectral radius $\rho(\mathbf{A}_{sys})$ is an important parameter that influences the decoding behavior. As shown in [34], when errors occur in the ETS, a larger spectral radius leads to slower error correction, and in some cases, it may even result in incorrect decoding. Therefore, we regard $\rho(\mathbf{A}_{sys})$ as a key indicator of the harmfulness of the ETS, with a larger spectral radius signifying a more harmful effect.

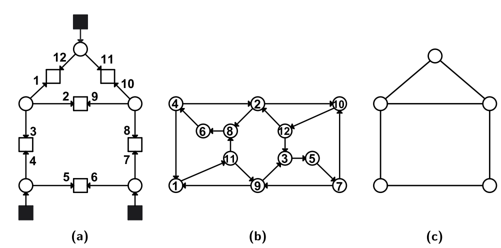

Fig. 2. (a) A $(5,3)$-ETS with labeled edges, where circles represent variable nodes, squares represent check nodes, and black solid squares represent 1-degree check nodes; (b) The update relationships of the directed edges in this ETS; (c) Corresponding VN graph.

For the reader's convenience, the parameters used in this paper are summarized in Table I.

III. Theoretical Results

In this section, we obtain results on Turán numbers and the spectral radius of the system matrix for various cycle lengths and different distances between cycles. Our theoretical analysis consistently shows that longer cycles and larger distances between cycles in the Tanner graphs lead to improved performance of the corresponding LDPC codes in the error floor region.

A. Turán Numbers of Two Classes of Graphs

We first determine the Turán numbers for the dumbbell graphs $D_b(3,3;0)$ and $D_b(3,3;1)$, in comparison with the result for the Turán number of $\theta(1,2,2)$ in [31]:

Theorem 3 [31]: For all $n \geq 4$, $ex(n,\theta(1,2,2)) = \lfloor \frac{n^2}{4} \rfloor$.

We begin by proving the following lemma, which is necessary for determining the Turán numbers of the two dumbbell graphs.

Lemma 1: Let n be a positive integer and $f : \mathbb{N} \to \mathbb{N}$ be a function satisfying the inequality $(n-2)(f(n)+1) - nf(n-1) > 0$. For a fixed graph H, if $ex(n-1, H) \le f(n-1)$, then $ex(n, H) \le f(n)$.

Proof: We prove this lemma by contradiction. Assume that there exists an H-free graph G with n vertices and exactly $f(n)+1$ edges.

We first prove that there exists a vertex v_0 in G with degree $d(v_0) \le f(n) - f(n-1)$. If not, for every vertex v in G, we have $d(v) \ge f(n) - f(n-1) + 1$. Consequently, the total number of edges satisfies

$$|E(G)| = \frac{1}{2}\sum_{v \in V(G)} d(v) \ge \frac{1}{2} n(f(n) - f(n-1) + 1).$$

On the other hand, we know that $(n-2)(f(n)+1) - nf(n-1) > 0$, which implies $f(n-1) < \frac{n-2}{n}(f(n)+1)$. Therefore, we obtain

$$\begin{aligned} |E(G)| &\ge \frac{1}{2} n(f(n) - f(n-1) + 1) \\ &> \frac{1}{2} n \left(f(n) - \frac{n-2}{n}(f(n)+1) + 1 \right) \\ &= f(n) + 1, \end{aligned}$$

However, this contradicts the assumption that $|E(G)| = f(n) + 1$. Therefore, there must exist a vertex v_0 with degree $d(v_0) \le f(n) - f(n-1)$.

Next, consider the graph $G - v_0$, where $|V(G - v_0)| = n-1$ and $|E(G-v_0)| \ge f(n-1)+1$. By assumption $ex(n-1, H) \le f(n-1)$, any graph with $n-1$ vertices and more than $f(n-1)$ edges must contain H. Since $|E(G - v_0)| \ge f(n-1) + 1$,

it follows that $G - v_0$, and thus G, must contain a copy of H. This contradicts the initial assumption that G is H-free. Then we have $ex(n, H) \le f(n)$.□

The following corollary can be easily proven by induction.

Corollary 1: In particular, if there exists $n_0 \in \mathbb{N}$ such that $ex(n_0, H) \le f(n_0)$ and the inequality $(n-2)(f(n)+1) - nf(n-1) > 0$ holds for all $n \ge n_0 + 1$, then for any $n \ge n_0$, we have $ex(n, H) \le f(n)$.

Based on Lemma 1 and Corollary 1, we can determine the exact values of $ex(n, D_b(3,3;0))$ and $ex(n, D_b(3,3;1))$.

Theorem 4: For all $n \ge 5$, $ex(n, D_b(3,3;0)) = \lfloor \frac{n^2}{4} \rfloor + 1$.

Proof: For the lower bound, consider a graph G_0 on n vertices with $E(G_0) = E\left(K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}\right) \cup \{xy\}$, where x and y are vertices not adjacent in $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$. It can be easily verified that G_0 does not contain a copy of $D_b(3,3;0)$, and thus, $ex(n, D_b(3,3;0)) \ge \lfloor \frac{n^2}{4} \rfloor + 1$. For the upper bound, let $f(n) = \lfloor \frac{n^2}{4} \rfloor + 1$. We now prove that $ex(n, D_b(3,3;0)) \le f(n)$.

For $n = 5$, consider a graph with $\lfloor \frac{n^2}{4} \rfloor + 2 = 8$ edges. Then it is either $K_5 - E(P_3)$ or $K_5 - E(2 \cdot P_2)$, both of which contain a copy of $D_b(3,3;0)$.

For $n = 6$, the inequality $(n-2)(f(n)+1) - nf(n-1) > 0$ holds, implying $ex(6, D_b(3,3;0)) \le \lfloor \frac{6^2}{4} \rfloor + 1 = 10$.

For $n = 7$, consider any graph G with $|E(G)| = f(n) + 1 = 14$. If there exists a vertex v' in G such that $d(v') \le 3$, then removing v' leaves $G - v'$ with 6 vertices and at least 11 edges. Given $ex(6, D_b(3,3;0)) = 10$, G must contain a copy of $D_b(3,3;0)$. If not, each vertex v in G has degree $d(v) \ge 4$, then with $|V(G)| = 7$ and $|E(G)| = 14$, G is 4-regular. Choose a vertex v_0 and denote its neighbors as $\{v_1, v_2, v_3, v_4\}$, with the remaining vertices being v_5 and v_6. Since $d(v_1) = 4$, then v_1 must be adjacent to at least one vertex in $\{v_2, v_3, v_4\}$, saying v_2. In the induced subgraph $G[\{v_3, v_4, v_5, v_6\}]$, there are at least $14 - 4 \times 3 + 3 = 5$ edges. By Theorem 1, there must be a C_3 in $G[\{v_3, v_4, v_5, v_6\}]$. If this C_3 is $\{v_3, v_4, v_5\}$ or $\{v_3, v_4, v_6\}$, then the induced subgraph of $\{v_0, v_1, v_2, v_3, v_4\}$ contains $D_b(3,3;0)$. If the C_3 is $\{v_3, v_5, v_6\}$ (or similarly $\{v_4, v_5, v_6\}$), since $d(v_3) = 4$, another neighbor of v_3, saying v_i, will result in induced subgraph of $\{v_0, v_i, v_3, v_5, v_6\}$ containing $D_b(3,3;0)$.

For $n \ge 8$:

- If $n = 2k$, $k \ge 4$,

$$\begin{aligned} &(n-2)(f(n)+1) - nf(n-1) \\ &= (2k-2)(k^2+2) - 2k(k^2-k+1) \\ &= 2k - 4 > 0. \end{aligned}$$

- If $n = 2k+1$, $k \ge 4$,

$$\begin{aligned} &(n-2)(f(n)+1) - nf(n-1) \\ &= (2k-1)(k^2+k+2) - (2k+1)(k^2+1) \\ &= k - 3 > 0. \end{aligned}$$

Thus, by Corollary 1, we have $ex(n, D_b(3,3;0)) \le \lfloor \frac{n^2}{4} \rfloor + 1$ for all $n \ge 8$. Combining the results for $n = 5$, $n = 6$, and $n = 7$, the proof is complete.□

Theorem 5:

$$ex(n, D_b(3,3;1)) = \begin{cases} 12, & \text{if } n = 6; \\ \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1, & \text{if } n \ge 7. \end{cases}$$

Proof: If $n = 6$, the complete bipartite graph $K_{3,3}$ with three additional edges $\{xy, yz, xz\}$ added within the same partition does not contain a $D_b(3,3;1)$. However, both $K_6 - E(P_3)$ and $K_6 - E(2 \cdot P_2)$ contain a $D_b(3,3;1)$. Therefore, $ex(6, D_b(3,3;1)) = 12$.

If $n \ge 7$, for the lower bound, consider a graph G_0 with $E(G_0) = E\left(K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}\right) \cup E\left(K_{1, \lceil \frac{n}{2} \rceil - 1}\right)$, where $K_{1, \lceil \frac{n}{2} \rceil - 1}$ is in the same part of $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$. One can verify that there is no $D_b(3,3;1)$ in G_0, so we conclude that $ex(n, D_b(3,3;1)) \ge \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1$. Now, we will prove $ex(n, D_b(3,3;1)) \le \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1$ for all $n \ge 7$.

For $n = 7$, consider a graph G_1 with 16 edges. If there exists a vertex t with degree at most 3, then $G_1 - t$ has 6 vertices and more than 12 edges, which guarantees the presence of a $D_b(3,3;1)$. If each vertex in G_1 has a degree of at least 4, the possible degree sequences are $\{6,6,4,4,4,4,4\}$ or $\{6,5,5,4,4,4,4\}$ or $\{5,5,5,5,4,4,4\}$. For these three degree sequences, we denote the corresponding vertices as $\{t_1, t_2, \ldots, t_7\}$. If the degree sequence is $\{6,6,4,4,4,4,4\}$ or $\{6,5,5,4,4,4,4\}$, by Pigeonhole principle, there exist two adjacent vertices t_i and t_j with $d(t_i) = d(t_j) = 4$. Then t_i, t_j and t_1 form a C_3, and by Theorem 1, there is a C_3 in $G_1 - \{t_1, t_i, t_j\}$. The two C_3 must be connected by an edge incident to t_1 as $d(t_1) = 6$, thus forming a $D_b(3,3;1)$. If the degree sequence is $\{5,5,5,5,4,4,4\}$, similarly, by Pigeonhole principle, there is a C_3 with vertices of degrees 5, 5 and 4, and the induced graph by the remaining vertices contains a C_3 by Theorem 1. Hence, we obtain a $D_b(3, 3; 1)$ and conclude that $ex(7, D_b(3, 3; 1)) \leq 15$.

For $n \geq 8$, define $f(n) = \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1$. For even $n = 2k$, consider a graph $G = (V, E)$ with $|V| = 2k$ and $|E| = f(2k) + 1 = k^2 + k$. If there exists a vertex v_0 with degree $d(v_0) \leq k$, then for the graph $G - v_0$ (with $2k - 1$ vertices), the number of edges satisfies $|E(G - v_0)| \geq k^2 = f(2k - 1) + 1$, implying the presence of a $D_b(3, 3; 1)$ in G. Otherwise, if each vertex has degree at least $k + 1$, then G is $(k + 1)$-regular, since $|E| = k^2 + k$. We claim that there is a $D_b(3, 3; 1)$ in this case, which, by assumption that $ex(2k - 1, D_b(3, 3; 1)) \leq f(2k - 1)$, implies $ex(2k, D_b(3, 3; 1)) \leq f(2k)$.

Fix a vertex v_0 in G with neighborhood $N(v_0) = \{v_1, v_2, \ldots, v_{k+1}\}$ and denote the remaining $k - 2$ vertices as $\{u_1, u_2, \ldots, u_{k-2}\}$. Note that v_1 must be adjacent to at least one vertex in $N(v_0)$, saying v_2. Consider the subgraph $G' = G - \{v_0, v_1, v_2\}$, which contains $2k - 3$ vertices and $E(G') = k(k + 1) - 3(k + 1) + 3 = k^2 - 2k \geq \lfloor \frac{(2k-3)^2}{4} \rfloor + 1$. By Theorem 1, there exists a C_3 in G'. Let $\{w_1, w_2, w_3\}$ be the vertices of this C_3. If any vertex in $\{w_1, w_2, w_3\}$ is adjacent to v_0, then the induced subgraph by $\{v_0, v_1, v_2, w_1, w_2, w_3\}$ contains a copy of $D_b(3, 3; 1)$. If none of w_1, w_2, w_3 is adjacent to v_0, then $\{w_1, w_2, w_3\} \subseteq \{u_1, u_2, \ldots, u_{k-2}\}$, and as $d(w_1) = k + 1$, w_1 must be adjacent to at least one vertex in $N(v_0)$. Suppose it is v_i. Similarly, v_i must be adjacent to some vertex v_j in $N(v_0)$ as $d(v_i) = k+1$. Hence, the induced subgraph by $\{v_0, v_i, v_j, w_1, w_2, w_3\}$ contains a copy of $D_b(3, 3; 1)$. Thus, there exists a $D_b(3, 3; 1)$ in G.

If $n = 2k + 1$, $k \geq 4$, we have $(n - 2)(f(n) + 1) - nf(n - 1) = (2k-1)\left(\lfloor \frac{(2k+1)^2}{4} \rfloor + \lceil \frac{2k+1}{2} \rceil\right) - (2k+1)\left(\lfloor \frac{(2k)^2}{4} \rfloor + \lceil \frac{2k}{2} \rceil - 1\right)$, which simplifies to $(2k - 1)(k^2 + 2k + 1) - (2k + 1)(k^2 + k - 1) = k > 0$. By Lemma 1, $ex(2k, D_b(3, 3; 1)) \leq f(2k)$ implies $ex(2k + 1, D_b(3, 3; 1)) \leq f(2k + 1)$. By induction, the proof is complete.□

For general dumbbell graphs with at least one odd cycle, we have:

Theorem 6: Let r_1, r_2, and q be positive integers, where $r_1 \geq 3$, $r_2 \geq 4$, $q \geq 0$, and $r_1 + r_2$ is odd. For $n \geq k^2 + k$ with $k = 3r_1 + 3r_2 + 2q - 12$, then $ex(n, D_b(r_1, r_2; q)) = \lfloor \frac{n^2}{4} \rfloor$.

Proof: The proof is given in Appendix A.□

Theorem 7: Let r_1, r_2, and q be positive integers, with r_1, r_2 odd, $3 \leq r_1 \leq r_2$, and $q \geq 0$. For $n \geq k^2 + k$ with $k = 3r_1 + 3r_2 + 2q - 11$, then

$$ex(n, D_b(r_1, r_2; q)) = \begin{cases} \lfloor \frac{n^2}{4} \rfloor + 1, & \text{when } q = 0; \\ \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1, & \text{when } q \geq 1. \end{cases}$$

Proof: The proof is given in Appendix B.□

In [37], the authors obtained the exact value of $ex(n, \theta(l_1, l_2, l_3))$ with at least one odd cycle.

Theorem 8 [37]: Let l_1, l_2, and l_3 be three positive integers with different parities, and assume $l_1 \leq l_2 \leq l_3$. For $n \geq 9k^2 - 3k$ with $k = l_1 + l_2 + l_3 - 1$, then $ex(n, \theta(l_1, l_2, l_3)) = \lfloor \frac{n^2}{4} \rfloor$.

Remark 2: In a Tanner graph with variable-regular degree γ, the VN graph of an (a, b)-ETS has a vertices and $\frac{1}{2}(a\gamma - b)$ edges. If H is forbidden in the VN graph, by the definition of the Turán number, we have $\frac{1}{2}(a\gamma - b) \leq ex(a, H)$, leading to the inequality $b \geq a\gamma - 2ex(a, H)$. Consequently, all (a, b)-ETSs with $b < a\gamma - 2ex(a, H)$ are eliminated. Therefore, for two fixed graphs H_1 and H_2 and a fixed positive integer a, if $ex(a, H_1) < ex(a, H_2)$, then $a\gamma - 2ex(a, H_1) > a\gamma - 2ex(a, H_2)$. This implies that removing H_1 from the VN graph can eliminate more (a, b)-ETSs than removing H_2.

For LDPC codes, (a, b)-ETSs with smaller values of a and b are more harmful to the performance in the error floor region. Additionally, by Remark 2, removing structures with smaller Turán numbers can eliminate more small ETSs, improving error floor performance. Furthermore, compared with Theorem 8, Theorem 6 and 7 indicate that dumbbell graphs (corresponding to non-negative cycle distances) generally have larger Turán numbers than theta graphs (corresponding to negative cycle distances). This implies that structures with smaller cycle distances are more critical to avoid.

Currently, constructing codes with larger girth (e.g., girth 8) is a more straightforward and effective method for eliminating many harmful ETSs. Our theorems theoretically suggest a potential strategy for improving the error floor by removing harmful subgraphs based on cycle distances. However, implementing this strategy in practice remains a challenge for future work.

In the next subsection, we analyze the spectral radius of the system matrix $\mathbf{A}_{sys}$ in the linear state-space model.

B. Spectral Radius

We first prove the theorem regarding the trend of the spectral radius of the system matrix when subdividing an edge in a graph. This result suggests that for ETSs, their harmful effect decreases as the cycle length or the distance between cycles increases. Subsequently, we calculate the spectral radii of the system matrices for several representative theta graphs and dumbbell graphs with varying cycle lengths and distances. The numerical results are consistent with our theoretical conclusions.

For a simple graph G, let $D(G) = \{(u, v), (v, u) \mid uv \in E(G)\}$. Regarding G as the VN graph of an ETS, we can define the system matrix $\mathbf{A}_{sys}(G)$ in two ways:

- Reconstruct the ETS from G and obtain $\mathbf{A}_{sys}(G)$ based on the ETS.
- Directly define $\mathbf{A}_{sys}(G)$ as a $|D(G)| \times |D(G)|$ matrix, where each row and column corresponds to an ordered pair in $D(G)$. Here, $\mathbf{A}_{sys}(G)(i, j) = 1$ if the second entry of j equals the first entry of i, and the first entry of j differs from the second entry of i; otherwise, $\mathbf{A}_{sys}(G)(i, j) = 0$.

Note that the two ways are equivalent. For example, the matrix $\mathbf{A}_{sys}$ in Example 1 is the system matrix of the VN graph shown in *Fig. 2 (c)*.

Lemma 2: Let G be a connected graph without any vertex of degree one. If G is not a cycle, then the system matrix $\mathbf{A}_{sys}(G)$ is irreducible.

Proof: See [34, Theorem 6].□

Theorem 9: Let G be a connected graph without any vertex of degree one, and suppose G is not a cycle. For an edge uv, denote the system matrix of G and G_{uv} as $\mathbf{A}_{sys}(G)$

and $\mathbf{A}_{sys}(G_{uv})$, respectively. Then we have $\rho(\mathbf{A}_{sys}(G_{uv})) \leq \rho(\mathbf{A}_{sys}(G))$.

Proof: By Lemma 2, both $\mathbf{A}_{sys}(G)$ and $\mathbf{A}_{sys}(G_{uv})$ are irreducible. We now show that there exists a vector $\boldsymbol{x}' \geq 0$ and $\boldsymbol{x}' \neq 0$ such that $\mathbf{A}_{sys}(G_{uv})\boldsymbol{x}' \leq \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'$. By Theorem 2 (ii), we can conclude $\rho(\mathbf{A}_{sys}(G_{uv})) \leq \rho(\mathbf{A}_{sys}(G))$.

Assume the eigenvector of $\mathbf{A}_{sys}(G)$ corresponding to $\rho(\mathbf{A}_{sys}(G))$ is $\boldsymbol{x}$, i.e., $\mathbf{A}_{sys}(G)\boldsymbol{x} = \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}$ and $\boldsymbol{x} \geq 0$. Denote the new vertex on the edge uv as w. We construct the vector $\boldsymbol{x}'$ as $\boldsymbol{x}'((u,w)) = \boldsymbol{x}'((w,v)) = \boldsymbol{x}((u,v))$, $\boldsymbol{x}'((v,w)) = \boldsymbol{x}'((w,u)) = \boldsymbol{x}((v,u))$, and $\boldsymbol{x}'(a) = \boldsymbol{x}(a)$ for other arcs $a \neq (u,w)$, (w,v), (v,w), and (w,u).

Denote $\boldsymbol{y}' = \mathbf{A}_{sys}(G_{uv})\boldsymbol{x}'$. We prove that $\boldsymbol{y}' \leq \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'$ by examining three cases:

- For any arc (v_i, v_j) with $v_i \neq u, v, w$,

$$\begin{aligned}\boldsymbol{y}'((v_i,v_j)) &= \sum_{(v_k,v_i)\in D(G_{uv}),v_k\neq v_j} \boldsymbol{x}'((v_k,v_i))\\ &= \sum_{(v_k,v_i)\in D(G),v_k\neq v_j} \boldsymbol{x}((v_k,v_i))\\ &= (\mathbf{A}_{sys}(G)\boldsymbol{x})((v_i,v_j))\\ &= \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((v_i,v_j)).\end{aligned}$$

- For any arc (u, v_j) with $v_j \neq w, v$,

$$\begin{aligned}\boldsymbol{y}'((u,v_j)) &= \sum_{\substack{(v_k,u)\in D(G_{uv})\\ v_k\neq v_j,w}} \boldsymbol{x}'((v_k,u)) + \boldsymbol{x}'((w,u))\\ &= \sum_{\substack{(v_k,u)\in D(G)\\ v_k\neq v_j,v}} \boldsymbol{x}((v_k,u)) + \boldsymbol{x}((v,u))\\ &= (\mathbf{A}_{sys}(G)\boldsymbol{x})((u,v_j))\\ &= \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((u,v_j)).\end{aligned}$$

  Similarly, we can obtain $\boldsymbol{y}'((v,v_j)) = \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((v,v_j))$ for any arc (v,v_j) with $v_j \neq u, w$.
- For the remaining four arcs (u,w), (w,u), (w,v), and (v,w), we have

$$\begin{aligned}\boldsymbol{y}'((u,w)) &= \sum_{(v_k,u)\in D(G_{uv}),v_k\neq w,v} \boldsymbol{x}'((v_k,u))\\ &= \sum_{(v_k,u)\in D(G),v_k\neq v} \boldsymbol{x}((v_k,u))\\ &= (\mathbf{A}_{sys}(G)\boldsymbol{x})((u,v))\\ &= \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((u,w)).\end{aligned}$$

  As there are no degree-one vertices in G, by Theorem 2 (iii), $\rho(\mathbf{A}_{sys}(G)) \geq 1$. Then we have

$$\begin{aligned}\boldsymbol{y}'((w,u)) &= (\mathbf{A}_{sys}(G_{uv})\boldsymbol{x}')((w,u)) = \boldsymbol{x}'((v,w))\\ &= \boldsymbol{x}'((w,u)) \leq \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((w,u)).\end{aligned}$$

  Similarly, $\boldsymbol{y}'((v,w)) = \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((v,w))$, $\boldsymbol{y}'((w,v)) \leq \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'((w,v))$.

Then we have $\boldsymbol{y}' \leq \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}'$ and finish the proof.□

Corollary 2: Let G be a connected graph with no vertex of degree one, and assume that G is not a cycle. If G is not regular, then for an edge uv, we have $\rho(\mathbf{A}_{sys}(G_{uv})) < \rho(\mathbf{A}_{sys}(G))$.

Proof: Since G is not regular and $\delta(G) \geq 2$, it follows from Theorem 2 (iii) that $\rho(\mathbf{A}_{sys}(G)) > 1$. Similar to the proof of Theorem 9, we can prove that there exists a vector $\boldsymbol{x} > 0$ such that $\mathbf{A}_{sys}(G_{uv})\boldsymbol{x} < \rho(\mathbf{A}_{sys}(G))\boldsymbol{x}$. Therefore, by Theorem 2 (ii), we conclude that $\rho(\mathbf{A}_{sys}(G_{uv})) < \rho(\mathbf{A}_{sys}(G))$.□

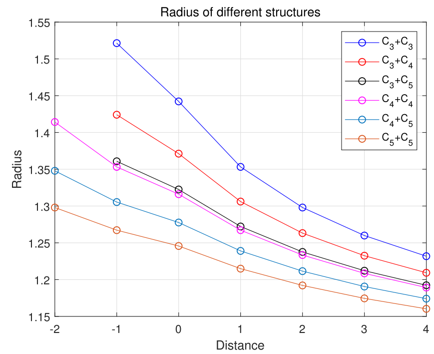

Fig. 3. The variation of the spectral radius with respect to different cycle lengths and distances between cycles.

Remark 3: The theta graph $\theta(l_1, l_2, l_3)$ satisfies the conditions in Corollary 2. Therefore, subdividing an edge uv reduces the spectral radius of the system matrix, which implies that $\rho(\mathbf{A}_{sys}(\theta(l_1,l_2,l_3))) > \rho(\mathbf{A}_{sys}(\theta(l_1+1,l_2,l_3)))$ (similarly for l_2 and l_3). Similarly, for the dumbbell graph $D_b(r_1,r_2;q)$, we have $\rho(\mathbf{A}_{sys}(D_b(r_1,r_2;q))) > \rho(\mathbf{A}_{sys}(D_b(r_1+1,r_2;q)))$ (similarly for r_2) and $\rho(\mathbf{A}_{sys}(D_b(r_1,r_2;q))) > \rho(\mathbf{A}_{sys}(D_b(r_1,r_2;q+1)))$ for $q \geq 1$.

We also compute specific examples for different cycle lengths and distances, with results summarized in Table II. The first column lists different distances between cycles, while the first row shows various combinations of cycle lengths. The symbol '-' indicates that no graph structure corresponds to the given distance and combination of cycles. For example, 'C_3, C_3' with a distance of '-1' corresponds to $\theta(1,2,2)$, and a distance of '1' corresponds to $D_b(3,3;1)$. Since the distance between two cycles of lengths c_1 and c_2 satisfies $d \geq \max\left\{-\left\lfloor\frac{c_1}{2}\right\rfloor, -\left\lfloor\frac{c_2}{2}\right\rfloor\right\}$, there is no graph corresponding to 'C_3, C_3' with a distance of '-2'.

The curve showing the variation of the spectral radius with different cycle lengths and distances is presented in Fig. 3, which provides a more intuitive visualization.

Observation 1: Based on Remark 3 and Fig. 3, we notice that longer cycles and larger distances between cycles lead to smaller spectral radii. This implies that such structures are less harmful to the performance of LDPC codes.

IV. The Impact of Distances Between Cycles on ETSs

In this section, we focus on the impact of the distance between two 6-cycles in the Tanner graph on ETSs, which corresponds to the relationship between two 3-cycles in the VN graph. We consider the cases where the distances between the two 3-cycles are -1, 0, and 1, corresponding to theta graph $\theta(1,2,2)$, dumbbell graph $D_b(3,3;0)$ and $D_b(3,3;1)$, respectively. We compare the harmfulness of these three structures from two perspectives. First, we use the Turán numbers of these graphs to calculate the ETSs eliminated when removing these structures. Second, we calculate the spectral radius of

TABLE II

THE SPECTRAL RADIUS AND THE CORRESPONDING GRAPHS FOR DIFFERENT CYCLE LENGTHS AND DISTANCES BETWEEN CYCLES

ρ	C_3, C_3	C_3, C_4	C_3, C_5	C_4, C_4	C_4, C_5	C_5, C_5
-2	-	-	-	1.4142 $\theta(2,2,2)$	1.3479 $\theta(2,2,3)$	1.2980 $\theta(2,3,3)$
-1	1.5214 $\theta(1,2,2)$	1.4241 $\theta(1,2,3)$	1.3608 $\theta(1,2,4)$	1.3532 $\theta(1,3,3)$	1.3054 $\theta(1,3,4)$	1.2672 $\theta(1,4,4)$
0	1.4422 $D_b(3,3;0)$	1.3712 $D_b(3,4;0)$	1.3225 $D_b(3,5;0)$	1.3161 $D_b(4,4;0)$	1.2776 $D_b(4,5;0)$	1.2457 $D_b(5,5;0)$
1	1.3532 $D_b(3,3;1)$	1.3061 $D_b(3,4;1)$	1.2722 $D_b(3,5;1)$	1.2672 $D_b(4,4;1)$	1.2390 $D_b(4,5;1)$	1.2149 $D_b(5,5;1)$
2	1.2980 $D_b(3,3;2)$	1.2632 $D_b(3,4;2)$	1.2376 $D_b(3,5;2)$	1.2334 $D_b(4,4;2)$	1.2115 $D_b(4,5;2)$	1.1921 $D_b(5,5;2)$
3	1.2599 $D_b(3,3;3)$	1.2325 $D_b(3,4;3)$	1.2121 $D_b(3,5;3)$	1.2085 $D_b(4,4;3)$	1.1906 $D_b(4,5;3)$	1.1745 $D_b(5,5;3)$
4	1.2318 $D_b(3,3;4)$	1.2093 $D_b(3,4;4)$	1.1932 $D_b(3,5;4)$	1.1892 $D_b(4,4;4)$	1.1741 $D_b(4,5;4)$	1.1603 $D_b(5,5;4)$

(a,b)-ETSs without these structures for various values of a and b. Furthermore, we use the linear state-space model to analyze the rate of error probability reduction of these ETSs as the number of iterations increases. Specifically, we use density evolution to approximate the information outside the ETSs and the linear state-space model to study the information updating process in the ETSs.

A. Turán Numbers of These Graphs and ETSs

By removing the theta graph $\theta(1,2,2)$ and the dumbbell graphs $D_b(3,3;0)$ and $D_b(3,3;1)$ in the VN graph of an (a,b)-ETS with variable degree γ, we obtain the following relationships among the parameters a, b, and γ, directly from Theorem 3, 4, and 5.

Corollary 3: For an (a,b)-ETS in a Tanner graph with variable-regular degree γ, the parameters a, b, and γ satisfy the following inequalities:

(1) If there is no $\theta(1,2,2)$ in the VN graph, then $b \geq a\gamma - \frac{1}{2}a^2$;

(2) If there is no $D_b(3,3;0)$ in the VN graph, then $b \geq a\gamma - \frac{1}{2}a^2 - 2$;

(3) If there is no $D_b(3,3;1)$ in the VN graph, then $b \geq a\gamma - \frac{1}{2}(a+1)^2 + 2$.

Based on Corollary 3, and considering the parity conditions of a, b, and γ, we provide the smallest values of a for various values of b when the above-mentioned theta graph and dumbbell graphs are removed from the VN graph of an (a,b)-ETS with variable-regular degree γ. These results are shown in Tables III. Computing the smallest value of a using the Turán numbers is less complicated compared to the computer enumeration method used in [30].

Based on the results presented, we notice that removing structures with smaller distance between cycles in a Tanner graph can eliminate more ETSs. From the perspective of Turán numbers, we provide a theoretical explanation for improving the performance of LDPC codes in the error floor region by ensuring a larger distance between any two cycles in the Tanner graph. This insight provides a feasible direction for the design of LDPC codes.

TABLE III

THE SMALLEST VALUE OF A FOR VARIOUS VALUES OF B WHEN THE CORRESPONDING THETA GRAPHS AND DUMBBELL GRAPHS ARE REMOVED FROM THE VN GRAPH

γ	b	$\theta(1,2,2)$-free	$D_b(3,3;0)$-free	$D_b(3,3;1)$-free
3	0	6	4	4
	1	7	5	5
	2	6	4	4
	3	5	3	3
4	0	8	8	5
	2	8	7	5
	4	7	6	4
	6	6	6	6
5	0	10	10	10
	1	11	11	9
	2	10	10	8
	3	11	9	9
	4	10	10	8
	5	9	9	7
	6	10	8	8
	7	9	9	7

B. Spectral Radius of ETSs

Based on Observation 1, structures with smaller distances between cycles are more harmful to the performance of LDPC codes, and we have shown that $\rho(\theta(1,2,2)) > \rho(D_b(3,3;0)) > \rho(D_b(3,3;1))$. In this subsection, we focus on the spectral radii of ETSs when these structures are forbidden. To isolate the impact of removing different structures, we calculate the spectral radius of the system matrix for each ETS in the following three disjoint sets and compute their median and mean values:

- $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$ represents all (a,b)-ETSs that contain $D_b(3,3;0)$ and $D_b(3,3;1)$ but are free of $\theta(1,2,2)$;

TABLE IV

THE NUMBER OF (a,b)-ETSS IN $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$ AND $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$, AND THE CORRESPONDING MEDIAN AND MEAN VALUES OF THE SPECTRAL RADIUS FOR VARIOUS VALUES OF A, B, AND $\gamma = 3$

Sets with $\gamma = 3$	Num.	ρ_{median}	ρ_{mean}
$\mathcal{S}_{\theta(1,2,2)-free}(6,2)$	1	1.6956	1.6956
$\mathcal{S}_{D_b(3,3;1)-free}(6,2)$	1	1.7293	1.7293
$\mathcal{S}_{\theta(1,2,2)-free}(7,3)$	1	1.5986	1.5986
$\mathcal{S}_{D_b(3,3;1)-free}(7,3)$	1	1.6653	1.6653
$\mathcal{S}_{\theta(1,2,2)-free}(8,2)$	3	1.7822	1.7921
$\mathcal{S}_{D_b(3,3;1)-free}(8,2)$	3	1.8250	1.8332
$\mathcal{S}_{\theta(1,2,2)-free}(8,4)$	2	1.5451	1.5451
$\mathcal{S}_{D_b(3,3;1)-free}(8,4)$	3	1.5990	1.6009
$\mathcal{S}_{\theta(1,2,2)-free}(9,1)$	3	1.9116	1.9132
$\mathcal{S}_{D_b(3,3;1)-free}(9,1)$	3	1.9189	1.9209
$\mathcal{S}_{\theta(1,2,2)-free}(9,3)$	7	1.7185	1.7226
$\mathcal{S}_{D_b(3,3;1)-free}(9,3)$	11	1.7471	1.7561
$\mathcal{S}_{\theta(1,2,2)-free}(10,2)$	15	1.8371	1.8469
$\mathcal{S}_{D_b(3,3;1)-free}(10,2)$	19	1.8461	1.8539
$\mathcal{S}_{\theta(1,2,2)-free}(10,4)$	18	1.6598	1.6880
$\mathcal{S}_{D_b(3,3;1)-free}(10,4)$	26	1.6916	1.7037

- $\mathcal{S}_{D_b(3,3;0)-free}(a,b)$ represents all (a,b)-ETSs that contain $\theta(1,2,2)$ and $D_b(3,3;1)$ but are free of $D_b(3,3;0)$;
- $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$ represents all (a,b)-ETSs that contain $\theta(1,2,2)$ and $D_b(3,3;0)$ but are free of $D_b(3,3;1)$.

We use the computational graph theory tool 'nauty' [38] to search for non-isomorphic VN graphs that satisfy our requirements. For each (a,b)-ETS with variable-regular degree γ, we search for an undirected graph on a vertices, with $\frac{1}{2}(a\gamma - b)$ edges and each vertex having a degree within the range $\left[\lceil\frac{\gamma}{2}\rceil, \gamma\right]$. We then count the number of the three graphs $\theta(1,2,2)$, $D_b(3,3;0)$, and $D_b(3,3;1)$ in each non-isomorphic VN graph and classify their corresponding ETSs into the three sets $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$, $\mathcal{S}_{D_b(3,3;0)-free}(a,b)$, and $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$ accordingly.

We calculate the spectral radius of (a,b)-ETSs with $\gamma = 3$ and 4 for all $a \leq 10$ and $b \leq 4$, as shown in Table IV and Table V. The following cases are omitted:

- For $\gamma = 3$, the dumbbell graph $D_b(3,3;0)$ is naturally avoided in all VN graphs. Thus, results for $\mathcal{S}_{D_b(3,3;0)-free}(a,b)$ are omitted. In this case, $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$ represents (a,b)-ETSs containing $D_b(3,3;1)$ but free of $\theta(1,2,2)$, while $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$ represents (a,b)-ETSs containing $\theta(1,2,2)$ but free of $D_b(3,3;1)$.
- Cases where $a\gamma - b$ is odd are omitted, as no such (a,b)-ETSs exist.
- Cases with $b = 0$ are omitted, as all graphs are regular and $\rho = \gamma - 1$ for all $(a,0)$-ETSs. This omission highlights the impact of removing different structures on the spectral radius.

In Table V, the '-' symbol indicates that the corresponding set is empty for certain values of a and b.

Observation 2: For all the cases listed in Table IV and V, we note that $\rho_{mean}(\mathcal{S}_{\theta(1,2,2)-free}(a,b)) < \rho_{mean}(\mathcal{S}_{D_b(3,3;0)-free}(a,b)) < \rho_{mean}(\mathcal{S}_{D_b(3,3;1)-free}(a,b))$. The same inequality holds for the median of the spectral radius. As mentioned in Section II, a larger spectral radius corresponds to a slower rate of error probability reduction. Therefore, removing $\theta(1,2,2)$ improves the performance of LDPC codes more than removing $D_b(3,3;0)$, and removing $D_b(3,3;0)$ is more effective than removing $D_b(3,3;1)$. In other words, $\theta(1,2,2)$ is more harmful to the performance than $D_b(3,3;0)$, and $D_b(3,3;0)$ is more harmful than $D_b(3,3;1)$.

TABLE V

THE NUMBER OF (a,b)-ETSS IN $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$, $\mathcal{S}_{D_b(3,3;0)-free}(a,b)$, AND $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$, AND THE CORRESPONDING MEDIAN AND MEAN VALUES OF THE SPECTRAL RADIUS FOR VARIOUS VALUES OF A, B, AND $\gamma = 4$

Sets with $\gamma = 4$	Num.	ρ_{median}	ρ_{mean}
$\mathcal{S}_{\theta(1,2,2)-free}(7,4)$	-	-	-
$\mathcal{S}_{D_b(3,3;0)-free}(7,4)$	2	2.4680	2.4680
$\mathcal{S}_{D_b(3,3;1)-free}(7,4)$	2	2.4745	2.4745
$\mathcal{S}_{\theta(1,2,2)-free}(8,2)$	-	-	-
$\mathcal{S}_{D_b(3,3;0)-free}(8,2)$	5	2.7777	2.7792
$\mathcal{S}_{D_b(3,3;1)-free}(8,2)$	1	2.7841	2.7841
$\mathcal{S}_{\theta(1,2,2)-free}(8,4)$	2	2.5348	2.5348
$\mathcal{S}_{D_b(3,3;0)-free}(8,4)$	11	2.5377	2.5365
$\mathcal{S}_{D_b(3,3;1)-free}(8,4)$	2	2.5488	2.5488
$\mathcal{S}_{\theta(1,2,2)-free}(9,2)$	4	2.8064	2.8061
$\mathcal{S}_{D_b(3,3;0)-free}(9,2)$	15	2.8074	2.8115
$\mathcal{S}_{D_b(3,3;1)-free}(9,2)$	2	2.8144	2.8144
$\mathcal{S}_{\theta(1,2,2)-free}(9,4)$	16	2.5918	2.5927
$\mathcal{S}_{D_b(3,3;0)-free}(9,4)$	52	2.5979	2.5984
$\mathcal{S}_{D_b(3,3;1)-free}(9,4)$	12	2.6180	2.6209
$\mathcal{S}_{\theta(1,2,2)-free}(10,2)$	39	2.8280	2.8290
$\mathcal{S}_{D_b(3,3;0)-free}(10,2)$	116	2.8291	2.8320
$\mathcal{S}_{D_b(3,3;1)-free}(10,2)$	6	2.8345	2.8353
$\mathcal{S}_{\theta(1,2,2)-free}(10,4)$	127	2.6393	2.6412
$\mathcal{S}_{D_b(3,3;0)-free}(10,4)$	387	2.6444	2.6454
$\mathcal{S}_{D_b(3,3;1)-free}(10,4)$	41	2.6582	2.6677

Furthermore, to more intuitively show the impact of removing different structures, we use the linear state-space model to calculate the average rate of error probability reduction in the ETSs of the three sets $\mathcal{S}_{\theta(1,2,2)-free}(a,b)$, $\mathcal{S}_{D_b(3,3;0)-free}(a,b)$, and $\mathcal{S}_{D_b(3,3;1)-free}(a,b)$ for the case of $(10,4)$-ETSs with $\gamma = 4$ as the number of iterations increases. For the external information from degree-1 check nodes, we use a Gaussian approximation for estimation, and we make the following assumptions, as done in [34] and [39]:

(1) The external inputs to the degree-2 check nodes in the ETS have a negligible effect on the information update within the ETS. This assumption holds because the Gaussian approximation for estimating external information grows rapidly, so the information transmitted by the degree-2 check nodes is largely dominated by the information within the ETS.

(2) The information generated within the ETS does not influence the external information updates. This is due to the fact that the number of nodes in the ETS is relatively small compared to the entire Tanner graph.

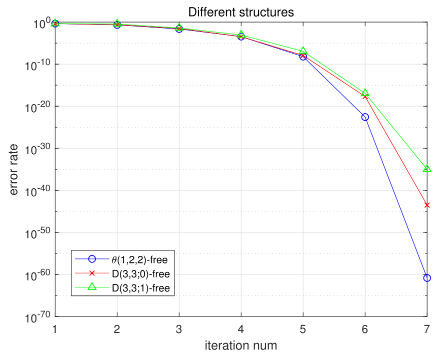

Fig. 4. The average error probability of ETSs in $\mathcal{S}_{\theta(1,2,2)-free}(10,4)$, $\mathcal{S}_{D_b(3,3;0)-free}(10,4)$, and $\mathcal{S}_{D_b(3,3;1)-free}(10,4)$ as the number of iterations increases for $\gamma = 4$.

(3) We use a Gaussian density function to approximate the probability density function of the messages. Additionally, we assume that the messages satisfy the consistency condition, meaning that the variance of the message's probability density function equals twice its mean. Thus, we only need to transmit the mean of the message during the process.

(4) We assume that the information transmitted from the channel and the degree-1 check nodes in each iteration is statistically independent. Moreover, we assume that $\boldsymbol{\lambda_{ex}}^{(l)}$ between iterations is also independent.

Based on these assumptions, we calculate the average error probability of $\mathcal{S}_{\theta(1,2,2)-free}(10,4)$, $\mathcal{S}_{D_b(3,3;0)-free}(10,4)$, and $\mathcal{S}_{D_b(3,3;1)-free}(10,4)$ for (10, 4)-ETSs with $\gamma = 4$ at iteration l:

$$Pr_e^{(l)}\{\mathcal{S}\} = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} Pr_e^{(l)}\{s\}, \tag{3}$$

where s represents an ETS in $\mathcal{S}$. For the error probability of an ETS, we estimate it by the maximum error probability of the variable nodes in the ETS:

$$Pr_e^{(l)}\{s\} = \max_{v\in s} Pr_e^{(l)}\{v < 0\}. \tag{4}$$

Based on assumption (3), we can rewrite equation (4) as

$$Pr_e^{(l)}\{s\} = \max_{v\in s} Q\left(\frac{\tilde{\lambda}^{(l)}(v)}{\sqrt{2|\tilde{\lambda}^{(l)}(v)|}}\right), \tag{5}$$

where $\tilde{\lambda}^{(l)}$ is obtained by the linear state-space model (2), with $Q(x)$ being the Gaussian Q-function.

We assume that the external Tanner graph corresponds to a (4,8)-regular LDPC code and that the channel variance is $\sigma = 0.83$. The messages from the degree-1 check nodes are characterized by a Gaussian approximation. For the variable nodes in the ETS, we assume that the mean of the channel information is 0.01, indicating very poor information from the channel. This assumption allows us to analyze the rate at which the error probability decreases within the ETS.

When $\gamma = 4$, as the number of iterations increases, the average error probabilities of the ETSs in $\mathcal{S}_{\theta(1,2,2)-free}(10,4)$, $\mathcal{S}_{D_b(3,3;0)-free}(10,4)$, and $\mathcal{S}_{D_b(3,3;1)-free}(10,4)$ are shown in Fig. 4, which further validate Observation 2.

V. Constructions and Numerical Results

Avoiding specific harmful trapping sets is an effective method for improving the performance of LDPC codes. Constructing QC-LDPC codes free of specific subgraphs (e.g., $D_b(3,3;0)$ and $D_b(3,3;1)$) requires a comprehensive analysis of all possible 6-cycles and their interconnections, based on Fossorier's equality [32]. However, deriving a concise and general formula to guarantee the absence of such structures is difficult. Given this challenge, we focus on a more general structural property: the distance between cycles. Instead of targeting specific graphs, we aim to systematically increase the cycle distances in the Tanner graph. This strategy is universally applicable to codes of arbitrary length and rate, and is effective for both regular and irregular constructions. Nevertheless, deriving precise exponent matrix conditions to avoid given subgraphs remains a valuable direction for future research.

Based on the insights from previous sections, we modify the classical progressive-edge-growth (PEG) algorithm [40]. The PEG algorithm operates by sequentially adding edges to the Tanner graph according to a prescribed variable node degree distribution. For each variable node being processed, the algorithm performs a breadth-first tree expansion and connects it to the reachable check node with the smallest degree in the deepest layer, thereby maximizing the local cycle length for the newly added edge. In contrast, our enhanced approach incorporates an additional optimization criterion: at each step of adding an edge, we not only maximize the local cycle length but also ensure that the newly formed cycles are as far as possible from the existing cycles in the Tanner graph.

To achieve this, we assign a weight $wt(v_i)$ to each variable node v_i to represent the cycles at v_i. Initially, the weights of all variable nodes are set to 0. Each time we perform a breadth-first expansion with v_i as the root, we retain all paths from v_i to each check node c_j, denoting these paths as $\mathcal{P}_{v_i}(c_j) = \{P \mid P \text{is a path from } v_i \text{to } c_j\}$. To save these paths, we check all connecting edges between adjacent layers in the expansion. Define the weight of c_j in this expansion as the sum of the weights of all variable nodes in $\mathcal{P}_{v_i}(c_j)$:

$$wt_{v_i}(c_j) = \sum_{P\in\mathcal{P}_{v_i}(c_j)} \sum_{\text{vaiable node } v\in P} wt(v) \tag{6}$$

Note that some variable nodes may appear multiple times in different paths from v_i to c_j in the expansion. Therefore, when calculating the weight of c_j, these variable nodes will be counted multiple times.

When the expansion of v_i does not include all check nodes, we connect v_i to the check node with the smallest degree that has not yet been included in the expansion, similar to the classical PEG algorithm. The difference arises when all check nodes have already been included in the expansion. In this case, we choose the check node c_j at the deepest layer with the smallest weight to connect. During this process, new cycles are formed, as each path from v_i to c_j creates a new cycle. Consequently, we update the weights of all variable nodes in $\mathcal{P}_{v_i}(c_j)$. Specifically, we add a fixed value w to the weight of each variable node. If a variable node appears k times in $\mathcal{P}_{v_i}(c_j)$, it indicates that k new cycles passing through this

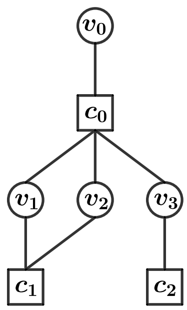

Fig. 5. An expansion with variable node v_0 as the root, where circles represent variable nodes and squares represent check nodes.

variable node are formed, so we add $k \times w$ to the weight of that variable node.

In this way, the weight of a variable node approximates the number of cycles passing through it, with a larger weight indicating more cycles. Therefore, we select the check node at the deepest layer with the smallest weight to connect. Choosing the deepest layer maximizes the local cycle length, while selecting the smallest weight ensures that the paths from the root to this check node intersect with the fewest cycles. This strategy guarantees that the newly formed cycle is as far as possible from existing cycles in the graph.

Note that the value of w can be adjusted to further distinguish between cycles of different lengths. In this paper, we set w as the reciprocal of the length of the newly formed cycle, to ensure that shorter cycles have larger weights than longer cycles. This prevents short cycles from being too dense in the Tanner graph. This process is further illustrated in the following example.

Example 2: For an expansion with v_0 as the root, as shown in Fig. 5, we have $\mathcal{P}_{v_0}(c_0) = \{(v_0c_0)\}$, $\mathcal{P}_{v_0}(c_1) = \{(v_0c_0v_1c_1), (v_0c_0v_2c_1)\}$, and $\mathcal{P}_{v_0}(c_2) = \{(v_0c_0v_3c_2)\}$. Then, $wt_{v_0}(c_1) = 2wt(v_0) + wt(v_1) + wt(v_2)$ and $wt_{v_0}(c_2) = wt(v_0) + wt(v_3)$. If $wt_{v_0}(c_1) < wt_{v_0}(c_2)$, we connect v_0 and c_1, forming 4-cycles, and set $w = \frac{1}{4}$. We then update the weights of the variable nodes as follows:

- The weight of v_0 is updated to $wt(v_0) + 2 \times \frac{1}{4}$;
- The weight of v_1 is updated to $wt(v_1) + \frac{1}{4}$;
- The weight of v_2 is updated to $wt(v_2) + \frac{1}{4}$.

We design the PEG-CYCLE algorithm.

For QC-LDPC codes, the expansion is performed with the first variable node of each QC block as the root, and the selection of connected check nodes follows the rules outlined in Algorithm 1. Based on this edge, all edges within the QC block are subsequently updated according to the quasi-cyclic structure. It is important to note that, due to the non-fully connected positions in the base matrix and the quasi-cyclic structure, some check nodes cannot be connected to the variable node. We refer to those that are able to add edges as available check nodes. After each edge addition, the available check nodes are updated. Furthermore, when updating the weights of each variable node, the same update must be applied to all variable nodes within the same QC block. We propose the QC-PEG-CYCLE algorithm.

Next, we present the performance curves of the QC-LDPC codes constructed using Algorithm 2. We compare our construction with the state-of-the-art constructions from [5], [41], [42], and [43], as well as the classical PEG algorithm [40], which includes both fully connected and non-fully connected constructions. In this section, all codes are decoded using the sum-product algorithm (SPA) over an additive white Gaussian noise (AWGN) channel, paired with binary phase shift keying (BPSK) modulation. The parameters of the simulated codes are given in Table VI.

Fig. 6 shows the performance of Algorithm 2 when applied to fully connected QC-LDPC codes. The codes C_1, C_2, and

Algorithm 1 PEG-CYCLE Algorithm

Require: Number of variable nodes n_v, number of check nodes n_c, degree sequence of variable nodes $\{d_1, d_2, \ldots, d_{n_v}\}$
Ensure: Parity-check matrix $\mathbf{H}$ of size $n_c \times n_v$

return $\mathbf{H}$ as a zero matrix, $wt(v) = 0$ for each variable node v
for $i = 1$ to n_v **do**
  for $k = 1$ to d_i **do**
    if $k = 1$ **then**
      Select the check node c_j with the smallest degree (random selection if multiple options), update $\mathbf{H}(j, i) \leftarrow 1$
    else
      Perform breadth-first expansion with v_i as the root, save paths $\mathcal{P}_{v_i}(c)$ for each check node c
      if there exist check nodes not included in the expansion **then**
        Select the check node c_j not yet in the expansion with the smallest degree (random selection if multiple options), update $\mathbf{H}(j, i) \leftarrow 1$
      else
        Select the check node c_j at the deepest layer with the smallest weight according to equation (6) (random selection if multiple options), update $\mathbf{H}(j, i) \leftarrow 1$
        for Each $P \in \mathcal{P}_{v_i}(c_j)$ **do**
          for Each $v \in P$ **do**
            Update $wt(v) \leftarrow wt(v) + w$
          end for
        end for
      end if
    end if
  end for
end for

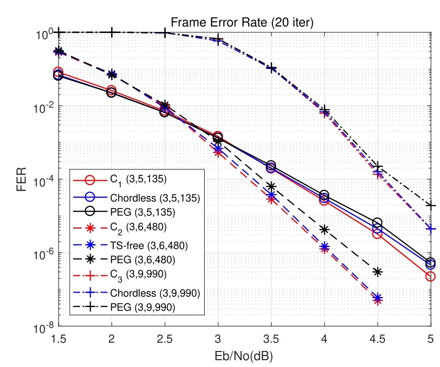

Fig. 6. FER performance of C_1, C_2, C_3, and their counterparts for the fully connected case under 20 iterations.

TABLE VI
THE PARAMETERS OF SIMULATED CODES

Code	Type	Lifting degree p	Length	Exponent matrix
C_1	(3,5)-regular	27	135	$\begin{bmatrix} 19 & 4 & 9 & 18 & 17 \\ 26 & 5 & 18 & 18 & 5 \\ 15 & 4 & 12 & 1 & 3 \end{bmatrix}$
Chordless [41]	(3,5)-regular	27	135	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 9 & 12 \\ 0 & 3 & 8 & 23 & 21 \end{bmatrix}$
C_2	(3,6)-regular	80	480	$\begin{bmatrix} 63 & 15 & 50 & 73 & 47 & 69 \\ 67 & 24 & 63 & 4 & 2 & 10 \\ 39 & 68 & 71 & 67 & 26 & 11 \end{bmatrix}$
TS-free [42]	(3,6)-regular	80	480	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7 & 39 & 41 & 45 & 61 \\ 0 & 35 & 43 & 51 & 66 & 36 \end{bmatrix}$
C_3	(3,9)-regular	110	990	$\begin{bmatrix} 85 & 7 & 94 & 83 & 11 & 107 & 72 & 35 & 72 \\ 94 & 93 & 50 & 71 & 17 & 7 & 86 & 1 & 4 \\ 19 & 34 & 79 & 19 & 93 & 107 & 87 & 66 & 78 \end{bmatrix}$
Chordless [41]	(3,9)-regular	110	990	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 8 & 12 & 22 & 46 & 63 & 78 \\ 0 & 2 & 7 & 18 & 28 & 15 & 76 & 98 & 51 \end{bmatrix}$
C_4	(4,8)-irregular	72	576	$\begin{bmatrix} 45 & \infty & \infty & 5 & 21 & 51 & 71 & 37 \\ 42 & \infty & 62 & \infty & 67 & 33 & 15 & 37 \\ \infty & 24 & 13 & \infty & 35 & 70 & 7 & 9 \\ \infty & 68 & \infty & 1 & 24 & 26 & 33 & 11 \end{bmatrix}$
Irregular TS-free [43]	(4,8)-irregular	72	576	$\begin{bmatrix} 0 & \infty & \infty & 0 & 0 & 0 & 0 & 0 \\ 0 & \infty & 4 & \infty & 13 & 30 & 33 & 52 \\ \infty & 17 & 5 & \infty & 39 & 12 & 52 & 47 \\ \infty & 58 & \infty & 64 & 6 & 1 & 24 & 29 \end{bmatrix}$
C_5	(4,10)-irregular	64	640	$\begin{bmatrix} 36 & 0 & 5 & 50 & \infty & \infty & 40 & \infty & \infty & 38 \\ 0 & \infty & \infty & 54 & 22 & 3 & 62 & 62 & 46 & 63 \\ 0 & 1 & \infty & 38 & 26 & \infty & \infty & \infty & 5 & \infty \\ \infty & 0 & 37 & \infty & 40 & 0 & 4 & 30 & 1 & 55 \end{bmatrix}$
5G BG2 [5]	(4,10)-irregular	64	640	$\begin{bmatrix} 9 & 53 & 12 & 26 & \infty & \infty & 61 & \infty & \infty & 13 \\ 39 & \infty & \infty & 38 & 61 & 61 & 34 & 28 & 32 & 60 \\ 17 & 50 & \infty & 44 & 52 & \infty & \infty & \infty & 48 & \infty \\ \infty & 8 & 58 & \infty & 60 & 40 & 17 & 54 & 18 & 0 \end{bmatrix}$
C_6	(4,12)-irregular	48	576	$\begin{bmatrix} \infty & \infty & 36 & 41 & 41 & 32 & 28 & 32 & 37 & 36 & 40 & 16 \\ \infty & 33 & \infty & 33 & 14 & 7 & 2 & 34 & 15 & 33 & 35 & 27 \\ 6 & 7 & \infty & \infty & 16 & 16 & 19 & 1 & 44 & 4 & 7 & 43 \\ 24 & \infty & 46 & \infty & 11 & 0 & 11 & 5 & 40 & 17 & 18 & 29 \end{bmatrix}$
Irregular TS-free [43]	(4,12)-irregular	48	576	$\begin{bmatrix} \infty & \infty & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \infty & 1 & \infty & 11 & 16 & 20 & 24 & 28 & 30 & 33 & 38 & 45 \\ 0 & 7 & \infty & \infty & 41 & 31 & 18 & 14 & 33 & 30 & 37 & 46 \\ 0 & \infty & 28 & \infty & 20 & 6 & 46 & 38 & 27 & 34 & 40 & 11 \end{bmatrix}$

C_3 are generated using Algorithm 2, and their performance is compared with the best-performing constructions from [41] and [42], denoted by 'Chordless' and 'TS-free', respectively. The constructions 'Chordless' are free of all cycles with a chord of length up to 12, while the construction 'TS-free' is free of all (a, b)-ETSs with $a \leq 12$ and $b \leq 3$. Our constructions perform similarly to, or even better than, the best codes from [41] and [42]. Furthermore, all four codes outperform those generated by the PEG algorithm.

The performance of non-fully connected codes is shown in Fig. 7. The codes C_4, C_5, and C_6 are generated using Algorithm 2, and their performance is compared with the best-performing constructions from [43] and the base matrix of 5G NR [5], denoted by 'Irregular TS-free' and '5G BG2', respectively. The constructions 'Irregular TS-free' focus on removing all (a, b)-ETSs with $a \leq 8$ and $b \leq 3$, while '5G BG2' corresponds to the case of $p = 64$ (without any shortening or puncturing). As shown in the figure, our constructions perform comparable to, or even superior to these state-of-the-art constructions. Moreover, all four codes outperform those generated by the PEG algorithm.

Algorithm 2 QC-PEG-CYCLE Algorithm

Require: Base matrix $\mathbf{B}$ of size $n_c \times n_v$, lifting degree p
Ensure: Parity-check matrix $\mathbf{H}$ of size $pn_c \times pn_v$

return $\mathbf{H}$ as a zero matrix, $wt(v) = 0$ for each variable node v, $\{d_1, d_2, \ldots, d_{n_v}\}$ is the degree sequence of $\mathbf{B}$
for $i = 0$ to $n_v - 1$ **do**
  for $k = 1$ to d_i **do**
    if $k = 1$ **then**
      Select an available check node c_j (random selection if multiple options), update $\mathbf{H}(j, 1 + ip) \leftarrow 1$
    else
      Perform breadth-first expansion with v_{1+ip} as the root, save paths $\mathcal{P}_{v_{1+ip}}(c)$ for each check node c
      if there exist available check nodes not included in the expansion **then**
        Select an available check node c_j not yet in the expansion (random selection if multiple options), update $\mathbf{H}(j, 1 + ip) \leftarrow 1$
      else
        Select the available check node c_j at the deepest layer with the smallest weight according to equation (6) (random selection if multiple options), update $\mathbf{H}(j, 1 + ip) \leftarrow 1$
        for Each $P \in \mathcal{P}_{v_i}(c_j)$ **do**
          for Each $v \in P$ **do**
            Update $wt(v) \leftarrow wt(v) + w$, update $wt(v') \leftarrow wt(v') + w$ for all variable nodes v' in the same QC block as v
          end for
        end for
      end if
    end if
    Update all edges in the same QC block with $(j, 1+ip)$, update the available check nodes
  end for
end for

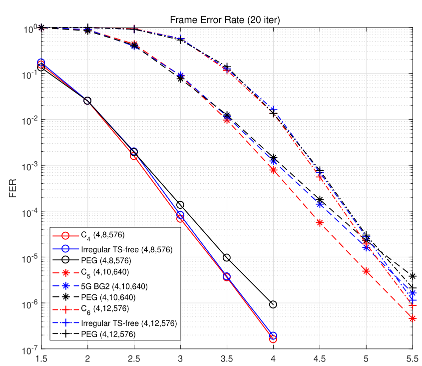

Fig. 7. FER performance of C_4, C_5, C_6, and their counterparts for the non-fully connected case under 20 iterations.

To evaluate the practical performance, we compare the performance of the aforementioned codes under 50 iterations, as shown in Fig. 8 and Fig. 9. At 50 iterations, the codes constructed using our algorithm outperform both the state-of-the-art construction methods and those generated by the PEG algorithm.

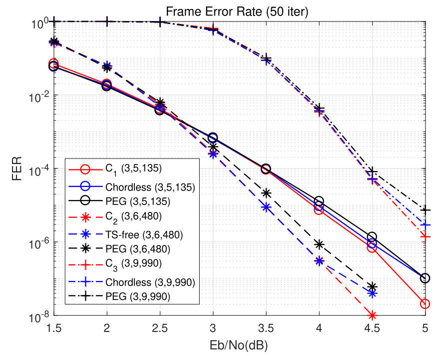

Fig. 8. FER performance of C_1, C_2, C_3, and their counterparts for the fully connected case under 50 iterations.

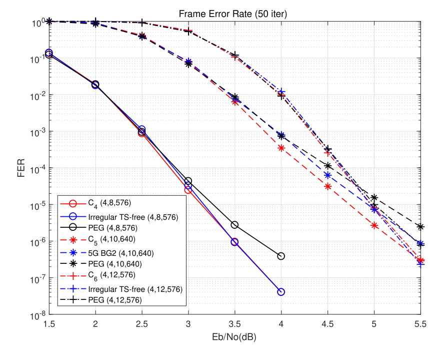

Fig. 9. FER performance of C_4, C_5, C_6, and their counterparts for the non-fully connected case under 50 iterations.

VI. Conclusion

In this paper, we study, for the first time, the impact of the distance between cycles on ETSs using theta graphs and dumbbell graphs. By determining two important graph parameters—the Turán numbers and spectral radii—for these graphs, we observe that removing structures with smaller distance between cycles can eliminate more ETSs in the Tanner graph. Moreover, a larger distance between cycles corresponds to a smaller spectral radius of the system matrix in linear state-space model, thereby improving the performance of LDPC codes.

We focus on three specific cases: two 6-cycles with distances of -1, 0, and 1 in the Tanner graph, represented by $\theta(1, 2, 2)$, $D_b(3, 3; 0)$, and $D_b(3, 3; 1)$ in the VN graph, respectively. Using their Turán numbers, we compute the ETSs eliminated by removing these graphs. Additionally, we analyze all $(10, 4)$-ETSs with $\gamma = 4$, calculating the average spectral radius and error probability for sets free of $\theta(1, 2, 2)$, $D_b(3, 3; 0)$, and $D_b(3, 3; 1)$. These results are consistent with our findings. We also design the PEG-CYCLE algorithm, which greedily maximizes the distance between cycles in the Tanner graph. The proposed algorithm shows improved performance for both fully connected and non-fully connected QC-LDPC codes. It outperforms the classical PEG algorithm and achieves performance comparable to, or even better than, state-of-the-art construction methods.

This work establishes a theoretical foundation for understanding the impact of cycle distances on ETSs. Based on

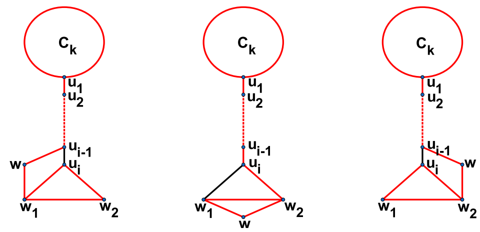

Fig. 10. A $D_b(3, k; i)$ in G labeled with the red color.

our findings, we suggest several directions for future research. Deriving necessary and sufficient conditions on the exponent matrix to avoid harmful structures such as $D_b(3, 3; 0)$ and $D_b(3, 3; 1)$ remains an open problem. Furthermore, exploring how to combine the principle of maximizing cycle distances with other construction constraints, such as maximizing girth or avoiding specific trapping sets, offers a viable approach for designing LDPC codes with superior error-floor performance.

Appendix A
Proof of Theorem 6

We first prove some necessary lemmas.

Lemma 3: Let t, k, and n be integers with $t \geq 2$, $k \geq 0$, and $n \geq \left\lfloor \frac{(t+1)^2}{4} \right\rfloor$. If G is a graph on n vertices and $|E(G)| \geq \left\lfloor \frac{n^2}{4} \right\rfloor + k$, then there exists an induced subgraph $G' \subseteq G$ on $n' \geq t$ vertices such that $|E(G')| \geq \left\lfloor \frac{n'^2}{4} \right\rfloor + k$ edges and $\delta(G') \geq \left\lfloor \frac{n'}{2} \right\rfloor$.

Proof: The proof follows a similar strategy to [37, Lemma 4]. For convenience, we outline the key steps here. The main idea of proof is iteratively deleting the vertex with the minimum degree.

Initially, let $G = G_0$. If $\delta(G_0) \geq \lfloor \frac{n}{2} \rfloor$, the proof is complete. Otherwise, there exists a vertex v_0 in G_0 with degree $d_{G_0}(v_0) \leq \lfloor \frac{n}{2} \rfloor - 1$. Let $G_1 = G_0 - v_0$, so G_1 contains $n-1$ vertices and satisfies $|E(G_1)| \geq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + k + 1$. If $\delta(G_1) \geq \lfloor \frac{n-1}{2} \rfloor$, the process terminates. Otherwise, we can find a vertex v_1 with $d_{G_1}(v_1) \leq \lfloor \frac{n-1}{2} \rfloor - 1$. Deleting v_1 from G_1 yields $G_2 = G_1 - v_1$, which has $n-2$ vertices and at least $\left\lfloor \frac{(n-2)^2}{4} \right\rfloor + k + 2$ edges. This procedure repeats until we obtain the desired graph $G_i = G'$ for some $i \leq n - t$. Otherwise, through this process, we construct a sequence of graphs $G_0, G_1, \ldots, G_{n-t}$ such that $|E(G_i)| \geq \left\lfloor \frac{(n-i)^2}{4} \right\rfloor + k + i$ and $\delta(G_i) \leq \lfloor \frac{n-i}{2} \rfloor - 1$. Consider G_{n-t}, which has t vertices and at least $\left\lfloor \frac{t^2}{4} \right\rfloor + k + n - t$ edges. This is greater or equal to $\frac{t(t-1)}{2} = |E(K_t)|$ when $n \geq \lfloor \frac{(t+1)^2}{4} \rfloor$, creating a contradiction with $\delta(G_{n-t}) \leq \lfloor \frac{t}{2} \rfloor - 1$. □

Next lemma allows us to extend the path of length 1 in $D_b(3, k; 1)$ to a path of length q:

Lemma 4: Given two fixed integers $q \geq 2$ and $k \geq 3$, let G be a graph on $n \geq 4k + 4q - 4$ vertices with $\delta(G) \geq \lfloor \frac{n}{2} \rfloor$. If G contains a $D_b(3, k; 1)$, then G also contains a copy of $D_b(3, k; i)$ for all $2 \leq i \leq q$.

Proof: We prove this lemma by induction. Suppose G contains a $D_b(3, k; i-1)$. We will show that G also contains a $D_b(3, k; i)$ for all $2 \leq i \leq q$.

Denote the vertices of $D_b(3, k; i-1)$ as $V(D_b(3, k; i-1)) = \{v_1, \ldots, v_{k-1}, u_1, \ldots, u_i, w_1, w_2\}$, where $\{v_1, v_2, \ldots, v_{k-1}, u_1\}$ forms the cycle C_k, $\{u_i, w_1, w_2\}$ forms the cycle C_3, and $\{u_1 u_2 \ldots u_i\}$ is a path of length $i-1$ connecting C_k and C_3. As

$$\begin{aligned}
&\sum_{x \in \{w_1, w_2, u_{i-1}\}} |N_G(x) - V(D_b(3, k; i-1))| \\
&\geq 3(\delta(G) - (k+i)) + 3 \\
&\geq 3\lfloor \frac{n}{2} \rfloor - 3(k+i) + 3 \\
&> n - (k+i+1) \\
&= |V(G) - V(D_b(3, k; i-1))|
\end{aligned}$$

by $n \geq 4k + 4q - 4 \geq 4k + 4i - 4$, the inequality shows that the total number of neighbors of w_1, w_2, and u_{i-1} outside $V(D_b(3, k; i-1))$ is strictly greater than the remaining vertices in the graph. By the pigeonhole principle, there must exist at least one vertex w in $V(G) - V(D_b(3, k; i-1))$ that is adjacent to at least two vertices from $\{w_1, w_2, u_{i-1}\}$. Once such a vertex w is found, we can add it to the existing structure of $D_b(3, k; i-1)$, thereby obtaining a $D_b(3, k; i)$ (see Fig. 10). □

Having extended the path in the dumbbell graph, we now use the following lemma from [37] to expand the cycles:

Lemma 5 [37]: Let l, m, n be three positive integers with $l \geq 3$, $m \geq 2$, and $n \geq 4l + 12m - 18$. Let G be a graph on n vertices with $\delta(G) \geq \lfloor \frac{n}{2} \rfloor$, and H be a subgraph of G with $|V(H)| = l$. If $u_0 w_1 v_0$ is a path in H, then G contains a (u_0, v_0)-path P of length $2m$ such that $V(P) \cap V(H) \subseteq \{u_0, w_1, v_0\}$.

We now obtain the following lemma, which extends the graph $D_b(x, y; q)$ to $D_b(r_1, r_2; q)$ where $r_1 \geq x$ and $r_2 \geq y$.

Lemma 6: Let x, y, r_1, r_2, q be five positive integers with $r_1 \geq x \geq 3$, $r_2 \geq y \geq 3$, $q \geq 0$, and r_1, x sharing the same parity, as well as r_2 and y. Let G be a graph on n vertices with $\delta(G) \geq \lfloor \frac{n}{2} \rfloor$. If $n \geq \min\{k_1, k_2\}$ and G contains a $D_b(x, y; q)$, where

$$\begin{aligned}
&k_1 \\
&= \max\{-2x+4y+4q+6r_1-10, -2y+4q+4r_1+6r_2-10\}, \\
&k_2 \\
&= \max\{-2y+4x+4q+6r_2-10, -2x+4q+4r_2+6r_1-10\},
\end{aligned}$$

then G also contains a $D_b(r_1, r_2; q)$.

Proof: We begin by applying Lemma 5 twice: the first application to the x-cycle and the second to the y-cycle in $D_b(x, y; q)$. This extension process yields an r_1-cycle and an r_2-cycle, respectively, resulting in a $D_b(r_1, r_2; q)$.

More specifically, denote

$$\begin{aligned}
&V(D_b(x, y; q)) \\
&= \{u_1, u_2, \ldots, u_x, w_1, w_2, \ldots, w_{q-1}, v_1, v_2, \ldots, v_y\},
\end{aligned}$$

where $\{u_1, u_2, \ldots, u_x\}$ forms the x-cycle, $\{v_1, v_2, \ldots, v_y\}$ forms the y-cycle, and the path of length q connecting these two cycles is $\{u_1, w_1, w_2, \ldots, w_{q-1}, v_1\}$.

In the case where $n \geq k_1$, let us denote $H_1 = D_b(x, y; q)$ and set $|V(H_1)| = l_1 = x + y + q - 1$, and $2m_1 = r_1 - x + 2$. Since $n \geq 4l_1 + 12m_1 - 18 = -2x + 4y + 4q + 6r_1 - 10$ and because $u_1 u_2 u_3$ is a path in H_1, Lemma 5 guarantees that G contains a (u_1, u_3)-path P_1 of length $2m_1 = r_1 - x + 2$ such that $V(P_1) \cap V(H_1) \subseteq \{u_1, u_2, u_3\}$. Therefore, $H_1 \cup P_1$ forms a $D_b(r_1, y; q)$, which we denote as H_2.

Similarly, we can set $2m_2 = r_2 - y + 2$, $|V(H_2)| = l_2 = r_1 + y + q - 1$, and apply Lemma 5 to the path $v_1 v_2 v_3$ in H_2.

Since $n \geq 4l_2 + 12m_2 - 18 = -2y + 4q + 4r_1 + 6r_2 - 10$, G must contain a (v_1, v_3)-path P_2 of length $2m_2 = r_2 - y + 2$ such that $V(P_2) \cap V(H_2) \subseteq \{v_1, v_2, v_3\}$. Then $H_2 \cup P_2$ yields a $D_b(r_1, r_2; q)$.

For the case where $n \geq k_2$, we first obtain a $D_b(x, r_2; q)$ from $D_b(x, y; q)$ and subsequently derive a $D_b(r_1, r_2; q)$. This completes the proof.□

We provide a brief overview of our proof framework: starting from the base cases of $D_b(3,3;0)$ and $D_b(3,3;1)$, we use Lemma 5 to extend the path in the dumbbell graph and use Lemma 6 to expand the cycles at both ends, thus completing the proof of the main theorem.

Note that the cycle expansion process maintains the parity of their lengths. This implies the necessity of using 3-cycles and 4-cycles to generate larger odd and even cycles, respectively. To utilize Lemma 6 for determining $ex(n, D_b(r_1, r_2; q))$ when $r_1 + r_2$ is odd, we need the following basic results for $D_b(3,4;0)$ and $D_b(3,4;1)$:

Lemma 7: For an integer $i \in \{0, 1\}$, let G be a graph on $n \geq 16 + 4i$ vertices and $\delta(G) \geq \lfloor \frac{n}{2} \rfloor$. If G contains a C_3, then G also contains a copy of $D_b(3, 4; i)$.

Proof: Denote the vertices of the C_3 as $\{v_1, v_2, v_3\}$. For $i = 0$, consider the neighbors of v_3. We aim to show that a C_4 exists in $V(G) - \{v_1, v_2\}$ that contains v_3. Given that $d_G(v_3) \geq \delta(G) \geq \lfloor \frac{n}{2} \rfloor$, we can choose $\{v_4, v_5, v_6\}$ from $N_G(v_3) - \{v_1, v_2\}$. Note that: $\sum_{x \in \{v_4, v_5, v_6\}} |N_G(x) - \{v_1, v_2, \ldots, v_6\}| \geq 3(\delta(G) - 5) + 3 \geq 3\lfloor \frac{n}{2} \rfloor - 12$. Since $n \geq 16 + 4i$, it follows: $3\lfloor \frac{n}{2} \rfloor - 12 > n - 6$. By the Pigeonhole principle, there must be a vertex v_7 in $V(G) - \{v_1, v_2, \ldots, v_6\}$ that is adjacent to at least two vertices from $\{v_4, v_5, v_6\}$, forming a $D_b(3, 4; 0)$.

For $i = 1$, apart from v_1 and v_2, we select another neighbor of v_3, denoted v_0, and then consider the neighbors of v_0. Similarly to the case of $i = 0$, we can find a C_4 in $V(G) - \{v_1, v_2, v_3\}$ that contains v_0 when $n \geq 16 + 4i$, leading the presence of a $D_b(3, 4; 1)$.□

Based on the foundational cases and the cycle expansion process, we can prove the following:

Proof of Theorem 6: For the complete bipartite graph $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$, it is obvious that there are no odd cycles, so $ex(n, D_b(r_1, r_2; q)) \geq |E(K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil})| = \lfloor \frac{n^2}{4} \rfloor$. Assume that G is a graph on $n \geq k^2 + k = \lfloor \frac{(2k+1)^2}{4} \rfloor$ vertices with $\lfloor \frac{n^2}{4} \rfloor + 1$ edges. By Lemma 3, there exists an induced subgraph G' on $n' \geq 2k = 6r_1 + 6r_2 + 4q - 24$ vertices with $\lfloor \frac{n'^2}{4} \rfloor + 1$ edges and $\delta(G') \geq \lfloor \frac{n'}{2} \rfloor$.

By Theorem 1, there exists a C_3 in G'. Applying Lemma 7, we find a $D_b(3, 4; i)$ in G' with $i \in \{0, 1\}$. For the case where $q = 0$, we use $D_b(3, 4; 0)$ to obtain $D_b(r_1, r_2; 0)$ through Lemma 6. For the case where $q > 0$, we apply Lemma 4 and Lemma 6 to extend the paths and cycles, respectively, finally obtaining $D_b(r_1, r_2; q)$ in G'. Therefore, we conclude that $ex(n, D_b(r_1, r_2; q)) \leq \lfloor \frac{n^2}{4} \rfloor$.□

Appendix B
Proof of Theorem 7

(i) When $q = 0$, to establish the lower bound, consider a graph G_0 on n vertices whose edge set $E(G_0) = E(K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}) \cup \{xy\}$, where x and y are non-adjacent vertices in $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$. Since each odd cycle in G_0 must contain the edge $e = xy$, it follows that no $D_b(r_1, r_2; 0)$ exists in G_0. Thus, we have $ex(n, D_b(r_1, r_2; 0)) \geq \lfloor \frac{n^2}{4} \rfloor + 1$.
For the upper bound, consider a graph G on n vertices with $|E(G)| \geq \lfloor \frac{n^2}{4} \rfloor + 2$. By Lemma 3, there exists an induced subgraph G' of G with $n' \geq 2k = 6r_1 + 6r_2 - 22$ vertices, $|E(G')| \geq \lfloor \frac{n'^2}{4} \rfloor + 2$ edges, and $\delta(G') \geq \lfloor \frac{n'}{2} \rfloor$. By Theorem 4, a $D_b(3, 3; 0)$ is found in G'. Then by Lemma 6, G' contains a $D_b(r_1, r_2; 0)$, as well as G. Therefore, we obtain $ex(n, D_b(r_1, r_2; 0)) \leq \lfloor \frac{n^2}{4} \rfloor + 1$.

(ii) When $q \geq 1$, consider the graph G_0 on n vertices with $E(G_0) = E(K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}) \cup E(K_{1, \lceil \frac{n}{2} \rceil - 1})$, where $K_{1, \lceil \frac{n}{2} \rceil - 1}$ is a star in the same part of $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$. Clearly, any two odd cycles in G_0 must share at least one common vertex, implying the absence of $D_b(r_1, r_2; q)$ in G_0. This leads to $ex(n, D_b(r_1, r_2; q)) \geq |E(G_0)| = \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1$.
For the upper bound, let G be a graph on $n \geq k^2 + k = \lfloor \frac{(2k+1)^2}{4} \rfloor$ vertices with $\lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil$ edges. By Lemma 3, there exists an induced subgraph G' on $n' \geq 2k = 6r_1 + 6r_2 + 4q - 22$ vertices, with $|E(G)| \geq \lfloor \frac{n'^2}{4} \rfloor + \lceil \frac{n}{2} \rceil \geq \lfloor \frac{n'^2}{4} \rfloor + \lceil \frac{n'}{2} \rceil$ edges, and $\delta(G') \geq \lfloor \frac{n'}{2} \rfloor$. By Theorem 5, there is a $D_b(3, 3; 1)$ in G'. By Lemma 4, we find a $D_b(3, 3; q)$ in G'. Through Lemma 6, G' contains a $D_b(r_1, r_2; q)$, as well as G. Thus, we have $ex(n, D_b(r_1, r_2; q)) \leq \lfloor \frac{n^2}{4} \rfloor + \lceil \frac{n}{2} \rceil - 1$.

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